# Key


Style 5

- Truth and Nil 6
- Arithmetic 16
- Constants and Chance 38
- Number Theory 49
- Polynomials and Roots 56
- Matrices 63
- Arrays 71
- Boxes 86
- Sets and Membership 97
- Trains 106
- Functions 124
- Errors 133
- Classes and Properties 144
- The Virtual 154

Dungeon 160

- Places 161
- Classes 165
- Objects 171
- Entities 176
- Food 179
- Species inheritance 198

Index 240

- English Index 241
- Glyph Index 248

# Contents


Style 5

Truth and Nil 6

Niladic 6

Monadic 8

Dyadic 10

Arithmetic 16

Niladic 16

Monadic 20

Dyadic 31

Constants and Chance 38

Niladic 38

Monadic 43

Dyadic 46

Number Theory 49

Monadic 49

Dyadic 52

Polynomials and Roots 56

Monadic 56

Dyadic 59

Matrices 63

Niladic 63
Monadic 65
Dyadic 68
Arrays 71
Niladic 71
Monadic 74
Dyadic 81
Boxes 86
Niladic 86
Monadic 88
Dyadic 92
Sets and Membership 97
Niladic 97
Monadic 99
Dyadic 102
Trains 106
Niladic 106
Monadic 109
Dyadic 121
Functions 124
Monadic 124
Dyadic 129
Errors 133
Niladic 133
Monadic 139
Classes and Properties 144
Niladic 144
Monadic 147

Dyadic 150
The Virtual 154
Niladic 154
Monadic 157
Dungeon 160
Places 161
Classes 165
Objects 171
Entities 176
Food 179
Species inheritance 198
Index 240
English Index 241
Glyph Index 248

# Style

# Truth and Nil: Niladic

## Wuxia

俠

*The Hero*

U+4FA0

## Nil



*The empty set — quiet failure value; absorbs; Nil⊡function → Nil*

U+2205

## Truth

⊤

*Verum — first-class truth; no true anymore*

U+22A4

# Truth and Nil: Monadic

## Invert

~

*Invert — monadic only (tilde). Scalar: ~ 0 → 1, ~ 1 → 0 (user wording to confirm). Unary scalar verbs operate on array back ranks (new 2026-10-11)*

U+007E

## Not

¬

*Not — monadic boolean not: ¬ Truth yields Nil, ¬ Nil yields Truth (new 2026-10-14)*

U+00AC

## Congruent

≡

*Ambivalent: monadic binds train to lhs⊡function*

U+2261

# Truth and Nil: Dyadic

## Less Than



*Less Than — dyadic < A B yields Truth when A < B, Nil otherwise (new 2026-10-14)*

U+003C

## Greater Than



*Greater Than — dyadic > A B yields Truth when A > B, Nil otherwise (new 2026-10-14)*

U+003E

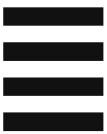
## Congruent



*Ambivalent: dyadic ≡ A B yields Truth when A and B have the same binding (same location in the system tables), Nil otherwise*

U+2261

## IsAt



*IsAt — dyadic ≣ X Y yields Truth when X and Y share the same pointer address, Nil otherwise (new 2026-10-14)*

U+2263

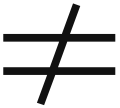
## Equality



*Equality — dyadic = A B yields Truth when A = B, Nil otherwise (new 2026-10-14)*

U+003D

## Unequal



*Unequal — dyadic ≠ A B: sugar for ¬ = A B (boolean Not of Equality) (new 2026-10-14)*

U+2260

## Less Or Equal



*Less Or Equal — dyadic ≤ A B yields Truth when A ≤ B, Nil otherwise (new 2026-10-14)*

U+2264

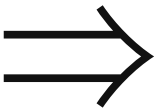
## Greater Or Equal



*Greater Or Equal — dyadic ≥ A B yields Truth when A ≥ B, Nil otherwise (new 2026-10-14)*

U+2265

## Implies



*Implies — logical implication; ∅ ⇒ ∅ → ⊤*

U+21D2

## forall



*Train quantifier — every element must evaluate non-nil. Idiom: dyadic NAND is ¬ ∀ 𝕀𝕀 X Y*

U+2200

## Approx



*Approx — dyadic ≈ A B yields Truth when|A - B|is within 1e-6, Nil otherwise (new 2026-10-14)*

U+2248

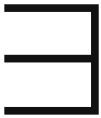
## UnApprox



*UnApprox — dyadic ≉ A B: sugar for ¬ ≈ A B (boolean Not of Approx) (new 2026-10-14)*

U+2249

## Forsome



*Existential — at least one element evaluates non-nil*

U+2203

## Fornone



*Not-forsome — defined as ~Forsome*

U+2204

# Arithmetic: Niladic

## Signum / Plus

+

*Ambivalent: Niladic + yields 0.*
*(Was ∓ U+2213, retired 2026-10-12*

U+002B

## Negate / Subtract

−

*Niladic - yields 0 (new 2026-10-11*

U+002D

## Plus (Complex)

⨢

*Plus (Complex), aka Sum Complex, Add Complex —; Niladic: sugar for ℿ 0 0, the complex zero (new 2026-10-14)*

U+2A22

## Times (Complex)

⨣

*Times (Complex) —; Niladic: sugar for ℿ 1 0, the complex one (new 2026-10-14)*

U+2A30

## Minus (Complex)



*Minus (Complex) —; Niladic: sugar for ⨿ 0 0, the complex zero (new 2026-10-14)*

U+235B

## Times (Polar)



*Times (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)*

U+2A02

## Divide (Complex)



*Divide (Complex) —; Niladic: sugar for ⨿ 1 0, the complex one (new 2026-10-14)*

U+2A2C

## Divide (Polar)



*Divide (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)*

U+2A38

## Factorial

!

*Factorial —; Niladic ! yields 1 (new 2026-10-12)*

U+0021

## Percent

%

*Percent —; Niladic % is 0 (new 2026-10-12*

U+0025

## Square



*Square — ⊠ X is X squared.; Niladic ⊠ yields 1, derived from Multiply (new 2026-10-12)*

U+22A0

## Multiply



*Multiply —; Niladic ⨯ yields empty array. Replaces Star * U+002A (new 2026-10-12; boxed forms 2026-10-12*

U+2A2F

# Arithmetic: Monadic

## Signum

+

*Ambivalent: monadic Signum (less than zero yields −1, greater than zero yields 1, zero yields 0; back-rank scalar unary)*

U+002B

## Negate

-

*Negate (monadic - X, unary minus)*

U+002D

## Conjugate

⊶

*Conjugate — monadic ⊶X: complex conjugate; negates the imaginary part, preserving orientation (on [r,i] yields [r,−i]; on [r;i] yields [r;−i]) (new 2026-10-15)*

U+22B6

## Double

+2

*Double — monadic +⍨X yields 2 × X (new 2026-10-15)*

U+2A27

## Sum

∑

*Sum — sugar for reduce-addition: ∑ X means / + X (redefined 2026-10-12)*

U+2211

## Square Root

√

*Square Root — monadic √ X yields the square root of X (new 2026-10-15)*

U+221A

## Plus (Complex)

⨁

*Plus (Complex), aka Sum Complex, Add Complex —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.*

U+2A22

## Times (Complex)

⨂

*Times (Complex) —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.*

U+2A30

## Minus (Complex)



*Minus (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.*

U+235B

## Times (Polar)



*Times (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.*

U+2A02

## Divide (Complex)



*Divide (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.*

U+2A2C

## Divide (Polar)



*Divide (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.*

U+2A38

## Percent



*Percent —; monadic % X is absolute value, a back-rank operator.; monadic redefined 2026-10-12)*

U+0025

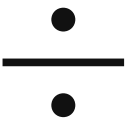
## Star



*Star — exponentiate: monadic * X is 1, because the missing exponent Y is assumed 0, giving X^0 (new 2026-10-12). Idiom: X-th root of Y is * Y ÷ 1 X*

U+002A

## Quotient



*Quotient —; Monadic ÷ X: if X is a square array, yields the matrix inverse; else reciprocal, a back-rank scalar operator (new 2026-10-12)*

U+00F7

## Multiply



*Multiply —; Monadic × X yields X when X numeric; on boxed X, catenates once.; monadic redefined 2026-10-12)*

U+2A2F

## Ceiling



*Ceiling —; monadic ⌈X rounds up*
*(from APL; new 2026-10-12)*

U+2308

## Floor



*Floor —; monadic ⌊X rounds down*
*(from APL; new 2026-10-12)*

U+230A

## Sine



*Sine (new 2026-10-12)*

U+25DD

## Cosine



*Cosine (new 2026-10-12)*

U+25DE

## Arccosine



*Arccosine (new 2026-10-12)*

U+25F6

## Arcsine



*Arcsine (new 2026-10-12)*

U+25F7

## Tangent



*Tangent (new 2026-10-12)*

U+2AD7

## Arctangent



*Arctangent (new 2026-10-12)*

U+2AD8

## Hyperbolic Sine



*Hyperbolic Sine (new 2026-10-12)*

U+2AA6

## Hyperbolic Cosine



*Hyperbolic Cosine (new 2026-10-12)*

U+2AA7

## Hyperbolic Arcsine



*Hyperbolic Arcsine (new 2026-10-12)*

U+2ABD

## Hyperbolic Tangent



*Hyperbolic Tangent (new 2026-10-12)*

U+2AD1

## Natural Log



*Natural Log —; monadic ⍁ X is ln X.; Monadic ⍁ on a polar 2×1 yields the rectangular complex pair [ln r, θ].*

U+237C

## Antilog



*Antilog — sugar for Star: ⍋̇ X is e raised to the X. Monadic ⍋̇ on a rectangular complex pair yields the polar 2×1 [e^a; b] (new 2026-10-12; glyph was 𝒆 U+1D58A until 2026-10-14)*

U+29CA

## Hyperbolic Arctangent



*Hyperbolic Arctangent (new 2026-10-12)*

U+2A4D

## Hyperbolic Arccosine



*Hyperbolic Arccosine (new 2026-10-12)*

U+2ABE

## Factorial

!

*Factorial —; monadic ! X is X factorial*

U+0021

## Fourier



*Fourier — monadic F X: FFT of X (new 2026-10-15)*

U+03DC

## InverseFourier



*InverseFourier — monadic ⅁ X: inverse FFT of X (new 2026-10-15)*

U+2132

## Base Ten Log



*Base Ten Log — ⍙ X is log base 10 of X, sugar for ⍦ 10 X (new 2026-10-12)*

U+2A5F

# Magnitude



*Magnitude — monadic ⍫X: absolute value of X (new 2026-10-16)*

U+2357

# Arithmetic: Dyadic

## Plus



*Ambivalent: dyadic + X Y yields X + Y (addition).; dyadic 2026-10-15)*

U+002B

## Subtract



*Subtract (dyadic - X Y yields X - Y, first minus second), the subtract that works with the / right-fold Reduce.; dyadic 2026-10-12)*

U+002D

## Plus (Complex)



*Plus (Complex), aka Sum Complex, Add Complex —; dyadic: sum of complex numbers (1×2 rectangular), per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before adding.*

U+2A22

## Times (Complex)



*Times (Complex) —; dyadic: product of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before multiplying.*

U+2A30

## Minus (Complex)



*Minus (Complex) —; dyadic: difference of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before subtracting.*

U+235B

## Times (Polar)



*Times (Polar) —; dyadic: product of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes multiply, angles add; if either or both X and Y are scalar, promote to polar 2×1 before multiplying.*

U+2A02

## Divide (Complex)



*Divide (Complex) —; dyadic: quotient of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before dividing.*

U+2A2C

## Divide (Polar)



*Divide (Polar) —; dyadic: quotient of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes divide, angles subtract; if either or both X and Y are scalar, promote to polar 2×1 before dividing.*

U+2A38

## Percent



*Percent —; dyadic % X Y yields the remainder of X divided by Y*

U+0025

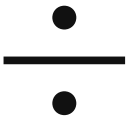
## Star



*Star — exponentiate: dyadic * X Y is X to the power Y (left=base, right=exponent)*

U+002A

## Quotient



*Quotient —; dyadic ÷ X Y divides X by Y, yielding a real; division by zero yields infinity, except 0/0 which yields Nil.*

U+00F7

## Multiply



*Multiply —; dyadic multiplies its arguments; when X is boxed, ⍪X Y catenates X repeated Y times.*

U+2A2F

## Factorial

!

*Factorial —; dyadic ! X Y is the binomial coefficient Y choose X.*

U+0021

## Ceiling

⌈

*Ceiling —; dyadic ⌈ X Y is maximum*

U+2308

## Floor

⌊

*Floor —; dyadic ⌊ X Y is minimum*

U+230A

## Natural Log



*Natural Log —; Dyadic ⍟ X Y is log base X of Y, sugar for ÷ ⍟ Y ⍟ X (new 2026-10-12)*

U+237C

# Square



*Square — ⊠ X is X squared.; Dyadic ⊠ X Y squares X Y times.*

U+22A0

# Constants and Chance: Niladic

## Half

½

*Half —; niladic: 0*

U+00BD

## Pi

π

*Pi — niladic π: value of pi (new 2026-10-14)*

U+03C0

## Bell

𝔅

*Bell —; niladic: random point in (-1, 1), exclusive*

U+1D56D

## Real

ℜ

*Real — niladic: random real number between 0 and 1; sugar for ½ + 1 𝔅 (new 2026-10-13)*

U+1D57D

## Gamma



*Gamma — niladic γ: Euler-Mascheroni constant (new 2026-10-14)*

U+03B3

## Golden



*Golden — niladic φ: golden mean (new 2026-10-14)*

U+03C6

## Euler



*Euler — niladic ℇ: Euler's constant (new 2026-10-14)*

U+2107

## Dice



*Dice —; niladic: single 6-sided die (1-6)*

U+1D56F

## Roll Percentile

‰

*Roll Percentile — niladic: a percentile roll (1-100); sugar for 𝅘𝅥𝅮100; the per mille sign U+2030 (2026-10-20)*

U+2030

## D2



*D2 — niladic: a single 2-sided die (1-2); sugar for 𝅘𝅥𝅮2; the die face U+2681 (2026-10-20)*

U+2681

## D4



*D4 — niladic: a single 4-sided die (1-4); sugar for 𝅘𝅥𝅮4; the die face U+2683 (2026-10-20)*

U+2683

## D6



*D6 — niladic: a single 6-sided die (1-6); sugar for 𝅘𝅥𝅮6; the die face U+2685 (2026-10-20)*

U+2685

# Pick



*Pick —; niladic: Nil*

U+2A03

# Constants and Chance: Monadic

## Half

½

*Half —; monadic ½ X: X divided by 2; array → sugar for ⊞(⍴ X); train → floor(length/2) index; dictionary → Nil (new 2026-10-13)*

U+00BD

## Pick

⍤

*Pick —; monadic ⍤X: scalar → same scalar; train → use Dice of size train to select element; array → use Dice of sides from shape to select element; dictionary → random {key:value} pair (new 2026-10-13)*

U+2A03

## Bell

𝔅

*Bell —; monadic 𝔅μ: bell curve point with mean μ (μ in (-1,1))*

U+1D56D

## Dice

𝔇

*Dice —; monadic 𝔇X: single X-sided die (1-X)*

U+1D56F

# Deal



*Deal — given a scalar, works as expected; given an array, fills the entire back rank with limited values. Concrete examples and summary TBD (new 2026-10-11)*

U+003F

# Constants and Chance: Dyadic

## Bell

𝔅

*Bell —; dyadic 𝔅 μ σ: bell curve point with mean μ and stddev σ (σ ≤ 1/3) (new 2026-10-13)*

U+1D56D

## Dice

𝔇

*Dice —; dyadic 𝔇 X N: N X-sided dice; all sugar via ℜ (hence 𝔅) (new 2026-10-13; die face U+2685 tried and reverted 2026-10-20)*

U+1D56F

# Number Theory: Monadic

## Divisible



*Divisible —; monadic ⍱ X yields Truth when X is divisible by 2, Nil otherwise*

U+23A2

## Indivisible



*Indivisible —; monadic ⍲ X: sugar for ¬ ⍱ X (the Odd test)*

U+2AEE

## Primes



*Primes — monadic 𝒫X yields the prime factors of X, per J's q:. (new 2026-10-18)*

U+1D4AB

## Prime



*Prime — monadic 𝔓 N yields the Nth prime, per J's p: (zero-indexed: 𝔓 0 is 2). (new 2026-10-18)*

U+1D513

## Prime



*Prime — monadic ⌽ X: sugar for ⍵⍵/ X ⍳2 ⌊√ X ⍸ ≤ X 1 (new 2026-10-15)*

U+23C0

## Factor



*Factor — factorization verb (new 2026-10-16)*

U+27D2

## Combinatoric



*Combinatoric —; monadic ## X: factorial of X*

U+2A69

# Number Theory: Dyadic

## LCM



*LCM — dyadic ⍜ X Y: least common multiple of X and Y (new 2026-10-16)*

U+238A

## Divisible



*Divisible —; dyadic ⍱ X Y yields Truth when X is divisible by Y, Nil otherwise (new 2026-10-15)*

U+23A2

## GCD



*GCD — dyadic ⫻ X Y: greatest common divisor of X and Y (new 2026-10-16)*

U+2980

## Indivisible



*Indivisible —; dyadic ⍲ X Y: sugar for ¬ ⍱ X Y (boolean Not of Divisible) (new 2026-10-15)*

U+2AEE

# Combinatoric



*Combinatoric —; dyadic ⋕ X Y: binomial X choose Y (new 2026-10-16)*

U+2A69

# Polynomials and Roots: Monadic

## NextLocal



*NextLocal — the Roots division step: given P, yields [R prime, P prime] where R prime is Local(P) and P prime is P deflated by synthetic division of R prime. (new 2026-10-19)*

U+2113

## Roots



*Roots — sugar: ℞ P is ℜ ℓ P (Reveal NextLocal over P); yields all polynomial roots. (new 2026-10-18; factored via Reveal 2026-10-19)*

U+211E

## Differentiate



*Differentiate — train verb ∂ T yields symbolically differentiated version of train T of coefficients, multiplying constants as needed; same order as Polynomial (new 2026-10-13)*

U+2202

## Integrate



*Integrate — train verb ∫ T yields symbolically integrated version of train T of coefficients, dividing constants as needed; same order as Polynomial (new 2026-10-13)*

U+222B

## Solve

&

*Solve — monadic & A: array → vector of polynomial coefficients; missing input row is all zeroes (new 2026-10-13)*

U+0026

## Optimize

𝔅

*Optimize —; monadic 𝔅 V: vector → polynomial; searches free variable with delta = 0.01 * domain, epsilon defaults to 1e-6*

U+214B

## Local

𝔏

*Local — monadic ⍵ V is 𝔅 ⫴𝔏 & V (FixedPoint Optimize Triple Solve V); approximates best x for a local minimal of V (new 2026-10-13)*

U+1D50F

# Polynomials and Roots: Dyadic

## Reveal



*Reveal — intrinsic. Dyadic ℜ P Y yields the successive application of P upon Y as Y shrinks to Nil. Epsilon from P has two elements: an alpha to apply P to next Y, and the next Y. (new 2026-10-18)*

U+211C

## Optimize



*Optimize —; dyadic 𝔅 V ε: recurse into smaller domains until change in c < ε (new 2026-10-13)*

U+214B

## Integer Division



*Integer Division — dyadic ⫽ X Y divides X by Y, yielding an integer (replaces Slash / integer division from manual p.19; / is now Reduce). Vector form: ⫽ V X where V is vector yields synthetic quotient of V by X (new 2026-10-14)*

U+2AFD

## Evaluate



*Evaluate — dyadic 𝒇X T evaluates train T of coefficients upon X (new 2026-10-13)*

U+1D58B

# FixedPoint



*Fixed point — dyadic ꞇø V guess: refine guess until epsilon < 1e-6 (max 100 steps); failure to converge returns infinity (new 2026-10-13)*

U+AB44

# Matrices: Niladic

## Outer Product



*Outer Product —; Niladic ⊗ yields 1 (new 2026-10-12)*

U+2297

## Inner Product



*Inner Product —; Niladic · yields 1 (new 2026-10-12; matrix case 2026-10-14)*

U+22C5

## Multiply Matrices



*Multiply Matrices —; Niladic ⋈ yields the 1×1 identity matrix (new 2026-10-14)*

U+22C8

# Matrices: Monadic

## Outer Product



*Outer Product —; Monadic ⊗ X yields X.*

U+2297

## Inner Product



*Inner Product —; Monadic ⋅ X yields X.*

U+22C5

## Multiply Matrices



*Multiply Matrices —; Monadic ⋈ X yields X.*

U+22C8

## L Factor



*L Factor — monadic: the L (lower triangular, 1s on diagonal) factor of the LU decomposition of a rank-2 array (new 2026-10-14)*

U+1D577

## Determinant



*Determinant — monadic ⌹X: determinant of square matrix X (new 2026-10-16)*

U+233C

## U Factor



*U Factor — monadic: the U (upper triangular) factor of the LU decomposition of a rank-2 array (new 2026-10-14)*

U+1D580

# Matrices: Dyadic

## Outer Product



*Outer Product —; dyadic ⊗ X Y yields the outer product matrix where entry [i,j] is ⍺ X[i] Y[j]; shape is the concatenation of the shapes of X and Y.*

U+2297

## Inner Product



*Inner Product —; dyadic ⋅ X Y: for vectors X, Y, the dot product (sum of pairwise products); for matrices, the Frobenius inner product: a scalar, the sum of all pairwise products.*

U+22C5

## Multiply Matrices



*Multiply Matrices —; dyadic ⋈ X Y: matrix multiplication, (⋈ X Y)[i;j] = ⋅ X[i;] ( ⌽ Y)[j;]; inner dimensions must agree, non-square fine.*

U+22C8

# Arrays: Niladic

Pee

ℙ

*Pee — Where⊡Pee yields Where [0, 1*
*(first two coordinates)*

U+2119

Zee

ℤ

*Zee — the last position of a slice*
*(Where⊡Zee yields Where [-1)*

U+2124

Exx

𝕏

*Exx — the zero position of a slice*
*(Where [0)*

U+1D54F

Wye

𝕐

*Wye — the midposition of a slice*

U+1D550

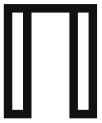
## Extend



*Extend —; niladic ⊢: sugar for ⊂ Nil*

U+213E

## Catenate



*Catenate —; niladic ⊣: sugar for ⊂ Nil*

U+213F

## Grade Up



*Grade Up — ⍋X yields the indices that sort X ascending. Niladic ⍋ yields 0 (from APL; new 2026-10-12)*

U+234B

## Grade Down



*Grade Down — ⍒X yields the indices that sort X descending. Niladic ⍒ yields Nil (from APL; new 2026-10-12)*

U+2352

# Arrays: Monadic

## Rank



*Rank — monadic # X yields the number of dimensions in X; zero when scalar. (new 2026-10-18)*

U+22D5

## Permute



*Permute —; monadic ⍨ X and dyadic X ⍨ Y work as in J's C.. Both modes intrinsic. (new 2026-10-19)*

U+2A0A

## Laminate



*Laminate —; monadic ⍪ X: adds a leading singleton axis*

U+2A5E

## Anagram



*Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)*

U+1D49C

## Enumerate

№

*Enumerate — monadic № X: sugar for ⍕⍳≢⍴X; yields index-value pairs for X (new 2026-10-18)*

U+2116

## Reflect

÷

*Reflect — monadic ÷X: sugar for ⍕⌽-⌽∘⍳X (new 2026-10-15)*

U+223B

## Count

77

⍳

U+2373

## Kount

⍳̂

*Kount — monadic ⍳̂B: takes a vector of T or Nil; returns a vector counting the T, turning each Nil to 0. Intrinsic. (new 2026-10-18)*

U+2A23

# ⍳ Count

U+2373

Count — magical: when given a shape, fills an array matching shape. Scalar: behaves as normal. Array: fills array with next dimension limited by scalar from back rank — e.g. ⍳ 2 3 → rows of 0 1 2 (origin 0; last axis counts, outer axes replicate) (new 2026-10-11)

## RavelRows



*RavelRows — monadic ⍪X: yields the rows of X as a vector (Antitable; new 2026-10-15)*

U+22F2

## Ravel



*Ravel — monadic ⍎X: flattens array X to vector, row-major (new 2026-10-14)*

U+22F3

## Reverse



*Reverse —; monadic ⌽ X reverses elements along the last axis*

U+233D

## Transpose



*Transpose — monadic ⍉ X reverses the order of axes (from APL; new 2026-10-12)*

U+2349

## Shape



*Shape —; monadic ρ X: shape of X; scalar → 0; array → vector of dimensions; train → [1, length of train]; dictionary → [1, pairs present]*

U+03C1

## Extend



*Extend —; monadic ⍐ X: Table X as in APL, adds trailing axis of length 1*

U+213E

## Catenate



*Catenate —; monadic ⍪ X: catenates X to itself*

U+213F

## Rotate



*Rotate —; monadic ⌽ X: rotate X toward origin by 1*

U+29BD

## Cardinality



*Cardinality — "how big is this?"; scalar → -1; array → number of elements held; dictionary → number of pairs; train → length of train (new 2026-10-13)*

U+2210

## Center



*Center — takes a shape, returns its integer halves expressed as a vector (e.g. [10, 6] → [5, 3]) (new 2026-10-13)*

U+229E

## Sight



*Sight — takes 2D array, finds center, returns vector location of nearest non-zero scalar; ties pick one (new 2026-10-13)*

U+2388

# Arrays: Dyadic

## Put



*Put — ⫤⊨ X P A: value at coordinate P of array A becomes X (new 2026-10-18)*

U+27DA

## Permute



*Permute —*

U+2A0A

## Laminate



*Laminate —; dyadic ⩢ A B: yields [A; B]; when A and B are both scalars, yields A B (new 2026-10-17)*

U+2A5E

## Anagram



*Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)*

U+1D49C

## Catenate



*Catenate —; dyadic ⫿ X Y: catenate X and Y as in APL; variadic ⫿ A ... Z: catenate A through Z left to right (new 2026-10-14)*

U+213F

## Take



*Take — ↑ X Y takes the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)*

U+2191

## Drop



*Drop — ↓ X Y drops the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)*

U+2193

## Reverse



*Reverse —; dyadic ⌽ X Y rotates Y by X positions (from APL; new 2026-10-12)*

U+233D

## Shape

ρ

*Shape —; dyadic ρ S X: reshape X to shape S (per APL) (new 2026-10-13)*

U+03C1

## Extend



*Extend —; dyadic ⍓ X N: extends by N axes (new 2026-10-14)*

U+213E

## Rotate



*Rotate —; dyadic ⌽̂ X Y: rotate X by Y (Y positive toward origin, negative away) (new 2026-10-16)*

U+29BD

# Boxes: Niladic

## Box



*Box —; Niladic ⊂ yields Truth*
*(from APL Enclose; new 2026-10-12)*

U+2282

## Unbox



*Unbox —; Niladic ⊃ yields Nil*
*(from APL Disclose; new 2026-10-12)*

U+2283

## Summarize



*Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; niladic ⊚ is the 1,1 vector containing zero.*

U+229A

# Boxes: Monadic

## Box



*Box — monadic ⍈ A: wraps A in a box so it travels as a single element (new 2026-10-17)*

U+2256

## Unbox



*Unbox — monadic ⍇ A: opens the box, yielding its contents (new 2026-10-17)*

U+2257

## Catalogue



*Catalogue —; monadic ⍀ X: cartesian product of the boxed lists in vector X (new 2026-10-17)*

U+2A37

## Raze



*Raze — monadic ⫻ A: sugar for Ravel Map Unbox A; opens every box and joins the contents into one vector (new 2026-10-17)*

U+2AF5

## Box



*Box —; monadic ⊂X encloses X into a scalar box.*

U+2282

## Unbox



*Unbox —; monadic ⊃X yields X.*

U+2283

## Summarize



*Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar*

U+229A

## Zip

91



U+22B0



# Zip

`U+22B0`

Zip — takes a train of vectors and zips them: ⍭ V1 V2 ... Vn yields an m×n matrix (m = vector length, n = train length); the resulting width is the length of the train; a scalar in the train expands into a column to fit the others; a container in the train contributes only its origin scalar, recursively (the summary effect); mismatched lengths Nil-pad (consistent with Diction); zips along the back rank for higher-rank arrays; ⍭ 1 2 yields the column [1;2] (new 2026-10-12; recycles the retired AntiSignum glyph)

# Boxes: Dyadic

## Summarize

94



U+229A

## Fetch



*Fetch — dyadic ⊸V X: applies V as search vector into X's items; follows path V down through nested boxes (new 2026-10-17)*

U+22B8

## Catalogue



*Catalogue —; dyadic ⊗A B: yields [[A0,B0] ... [A1,B1]] ... [[AN,B0] ... [AN,BN]], the outer product matrix of pairs*

U+2A37

## Link



*Link — dyadic ⫢X Y: sugar for Catenate Map Box X Y; boxes X and Y into a two-element list (new 2026-10-17)*

U+2AE9



# Summarize

U+229A

Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; Dyadic D ⊚ A: D (Direction) is a rank-1 array of coordinates; summarization starts at A[D], then proceeds as monadic; a scalar Direction s means start at (s, 0); Zip summarizes each train element (new 2026-10-12)

## Box



*Box —; Dyadic ⊂ X Y encloses X by Y levels; Y=0 leaves X unchanged.*

U+2282

## Unbox

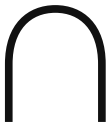


*Unbox —; Dyadic ⊃ X Y picks element X from Y.*

U+2283

# Sets and Membership: Niladic

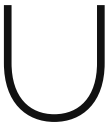
## Intersection



*Intersection —; niladic ∩: sugar for ⊂ Nil*

U+2229

## Union

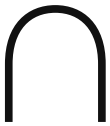


*Union —; niladic ∪: sugar for ⊂ Nil*

U+222A

# Sets and Membership: Monadic

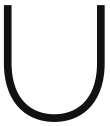
## Intersection



*Intersection —; monadic ∩ A: is A*

U+2229

## Union



*Union —; monadic ∪X: unique as in APL*

U+222A

## InOf



*InOf — the curry of In: ⇔InOf ⍷∈ X; monadic predicate testing membership in X. (new 2026-10-18)*

U+22F8

## Where



*Where — monadic ⍸Y: returns vector of indices of all elements of array Y that are Truth (new 2026-10-18)*

U+2A42

# Enlist

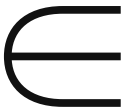


*Enlist —; monadic ∈Y: as defined in APL (deep flatten to vector)*

U+220A

# Sets and Membership: Dyadic

## In



*In — dyadic ∈ Y X yields Truth when X is in Y, Nil otherwise (J's e.). (new 2026-10-18)*

U+2208

## Bucket



*Bucket — dyadic ∐ X Y: yields index of interval in X that Y falls into (new 2026-10-18)*

U+2210

## InLast



*InLast — dyadic ∈̇ X Y: yields the index of the last occurrence of X in Y; yields length of Y if not found (new 2026-10-18)*

U+22F5

## InAt



*InAt — dyadic ⋲ X Y: yields the index of Y whose value equals X; yields length of Y if not found (new 2026-10-18)*

U+22F9

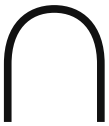
## In



*Enlist —; dyadic ∈ X Y: find/membership as defined in APL (new 2026-10-13)*

U+220A

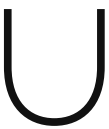
## Intersection



*Intersection —; dyadic ∩ A B: as in APL, elements in both (new 2026-10-14)*

U+2229

## Union



*Union —; dyadic ∪ X Y: union as in APL (new 2026-10-14)*

U+222A

## Set Difference



*Set Difference — dyadic ⍠A B: elements in A but not in B (new 2026-10-14)*

U+2A43

# Trains: Niladic

## Reduce



*Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Niladic / returns 0 (new 2026-10-12)*

U+002F

## Map



*Map — ⍬ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Niladic ⍬ yields Truth, same as niladic 𝔗 (new 2026-10-12)*

U+226C

## Reduce (forward)



*⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.*

U+233F

## Identity



*Identity — 𝔗 V yields V unchanged; on a train, yields the train unchanged. Niladic 𝔗 yields Truth (new 2026-10-12)*

U+1D574

## Backstroke



*Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Niladic \ yields 0 (new 2026-10-12)*

U+005C

## Stop



*Stop — statement separator; separates statements on one line*

U+22C4

## Backstroke (forward)



*Backstroke (forward) — Reverse Scan, paired with ⫻. ⫲ F V on a length-n V yields a length-n vector where element i is ⫻F (left fold) applied to the first i elements of V.*

U+2340

## Strip



*Strip —; niladic: removes the train from head up to but not including the first Stop encountered*

U+2ACF

# Trains: Monadic

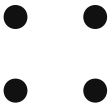
## Meter

111



U+2214

## Atomize



*Atomize — monadic ∴A: reads up to the first 32 elements of A (origin to end) and prepares them for use by train elements ahead (new 2026-10-14)*

U+2237

## Duplicate



*Duplicate — copies a train to a train twice as long: ⊷A B C → A B C A B C (was ==U+2A75, then ∴ U+2236 — settled 2026-10-12)*

U+22B7

## Caboose



*Caboose — picks up the train from head to Stop, then converts boxed contents to a vector (inverse of Atomize) (new 2026-10-15)*

U+2A78



# Meter

U+2214

Meter — measures its argument: ∔ V is ≺ + ⊷ V (zip the needle with the signal); yields M, a sequence of (needle, signal) pairs, needle ∈ -/0/+. M works in place of V in all Collate modes: the needle selects the function, the function transforms the signal (was ±, corrected to ≺ 2026-10-12)

## Reduce



*Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Monadic / V yields a linear array of the scalars ripped from V, highest to lowest indices.*

U+002F

## Map



*Map — ⍁ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Monadic ⍁ B is identical to Identity 𝔗 B.*

U+226C

## Reduce (forward)



*⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.; Monadic ⌿ V and niladic ⌿ behave identically to / (new 2026-10-12)*

U+233F

## Zero



*Zero — on a train, replaces all values with zeroes while keeping shapes (new 2026-10-12)*

U+1D585

## Backstroke

114

\

U+005C

## Duplicate Head

Ⅱ

*Duplicate Head — Ⅱ A ... yields A A ...; duplicates head of train, aka duplicate/dup (new 2026-10-14)*

U+2161

## Backstroke (forward)

⧵

*Backstroke (forward) — Reverse Scan, paired with ⫻. ⧵ F V on a length-n V yields a length-n vector where element i is ⫻F (left fold) applied to the first i elements of V.; Monadic ⧵ V and niladic ⧵ behave identically to \ (new 2026-10-12)*

U+2340

## Pipeline



*Pipeline —*

U+2AF7



# Backstroke

`U+005C`

Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Monadic \ V yields a linear array of the scalars ripped from V, lowest to highest indices.

## Head



*Head — sugar for ⊐1 (new 2026-10-15)*

U+2991

## Tail



*Tail — sugar for ⊐-2 (new 2026-10-15)*

U+2992

## Duplicate Tail



*Duplicate Tail — ⧟ ... A yields ... A A; duplicates tail of train (new 2026-10-13)*

U+29DE

## Strip

116



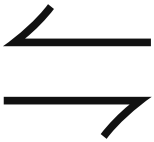
U+2ACF

# ⊐ Strip

U+2ACF

Strip —; monadic ⊐ X: removes the train from position X (0 is head) up to but not including the first Stop encountered; when X < 0, strips from position X in reverse order back to head, where −1 is the final tail position adjacent to the first Stop encountered (new 2026-10-15)

## Swap



*Swap — ⇋A B ... yields B A ...; swaps head two of train. Single: ⇋ A Stop X yields X. Bare: ⇋ Stop X yields X (new 2026-10-13)*

U+21CB

## Replicate

118



U+232C

## Rotate Head



*Rotate Head — ⟲A B C ... yields B C ... A; rotates head to tail (new 2026-10-13)*

U+27F2

## Rotate Tail



*Rotate Tail — ⟳ ... A B C yields C ... A B; rotates tail to head (new 2026-10-13)*

U+27F3



# Replicate

U+232C

Replicate — ⌬ N X Stop yields X replicated N times; e.g. ⌬ 5 A Stop gives A A A A A. Zero case: ⌬ 0 A Stop X yields X. Negative N is antireplicate: ⌬ -1 A A A A A Stop yields A A A A. Empty antireplicate: ⌬ -1 Stop X yields X. Bare: ⌬ Stop X yields X (new 2026-10-13)

## Nip



*Nip — ↢ A B ... yields A ...; drops second, keeps head (new 2026-10-13)*

U+21F7

## Tuck



*Tuck — ↣ A B ... yields A B A ...; copies head under second (new 2026-10-13)*

U+21F8

## Reverse Train



*Reverse Train — reverses order of train (new 2026-10-13)*

U+21F9

## Over



*Over — ⤺ A B ... yields B A B ...; copies second to head (new 2026-10-13)*

U+293A

## Triple

Ⅲ

*Triple — Ⅲ X yields X X X; triplicates top of stack (new 2026-10-13)*

U+2162

## Fork



*Fork — ⋔ F G H X: sugar for G H Swap F Dup X (J fork; new 2026-10-14)*

U+22D4

## Drop Train



*Drop Train — sugar for reverse, replicate -1, reverse; drops head of train (new 2026-10-13)*

U+291A

# Trains: Dyadic

## Applyself



*Applyself — ⍣⍤1 f n a applies f n times to a; sugar for ⫷a ⍵⊂⍣⍺⍵ f n (from APL power; new 2026-10-12)*

U+2363

## Pipeline



*Pipeline —; dyadic ⫷X Y applies vector Y, readable as a train of monadic functions, in sequence to X (new 2026-10-12)*

U+2AF7

# Functions: Monadic

## Diction

¡

*Diction — dict-backed decision-table constructor (was Vectorize, renamed 2026-10-11). ¡ A B → {A:B}; ¡ A B C D → {A:B,C:D}; monadic ¡ (a b c) self-maps into a set. Membership sugar: X member Y → ⊤(X ¿ (¡ Y))*

U+00A1

## Glyph

®

*Glyph — the name for a single character value. Glyph ← ≡ ⌘: Glyph's function is the literalize train (via monadic ≡) — to make a Glyph, literalize*

U+00AE

## Decide

126

¿

U+00BF

## Function

∇

*Function — gets glyph del (∇, nabla): the function-definition operator. Monadic ∇Y executes array Y as an FSA. (monadic 2026-10-19)*

U+2207

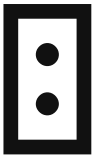


# Decide

U+00BF

Decide — first-match generalized switch/match over n×2k train tables (was When, 2026-10-11). n×2 returns 1×1 (scalar); n×4 returns 1×2; generally n×2k returns 1×k. An explicit trailing C is unmatched-result filler, not an else branch; without filler, unmatched returns Nil without logging an error. Canonical: R ← X ¿ ¡ A B C

## Print



*Print — the quad-colon verb: prints the value from its right, then passes that value to its left (new 2026-10-11)*

U+2360

## (literalize)



*Literalize next nonspace char, or ※0x???? by codepoint*

U+2A33

## Rail



*Rail — monadic ⌐¯X yields a vector of symbols from train X (train-to-array conversion). (new 2026-10-19)*

U+2ACD

## Lay



*Lay — monadic ¯¬V yields the train laid down from origin to limit of vector V (array-to-train, inverse of Rail). (new 2026-10-19)*

U+2ACE

# Lambda

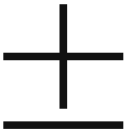


*Lambda — lambda abstraction (new 2026-10-13)*

U+039B

# Functions: Dyadic

## Collate



*Collate — executes one of three trains upon a value: for less than zero, for zero, or for greater than zero (was AntiSignum ≺ U+22B0, renamed 2026-10-12; complement to signum)*

U+00B1

## Curry

131



U+1D49E



# Curry

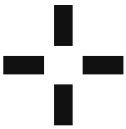
U+1D49E

Curry — intrinsic. Partial application: turns a dyadic verb into a monadic predicate by binding an argument, e.g. making = X from =. Semantics: Gets Q Curry P args: args to omega prime, omega prime shifts into alpha prime (binding the leftmost argument, textbook curry order), Q X makes X alpha, gets slice -1 of omega prime A, then call P; Q alpha yields P evaluated upon alpha prime, omega prime, and alpha as arguments in that order. Omega becomes nil upon Q's execution. Principle: Gets Q Curry P train is the principle for creating curries (Q is an example name assigned to the curried function, not a verb). (new 2026-10-18)

# Errors: Niladic

## Silence

135



U+22B9

## Compress

136



U+22D6

## Errline



*Last error: which line*

U+274C

## Errmsg



*Last error: what (type)*

U+274E



# Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Niladic ⊹ yields boxed Nil, a trial vector for enclosing operators to work on for starts (the dual of Compress; new 2026-10-12)



# Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Niladic ⋖ yields boxed Truth, a trial vector for enclosing operators to work on for starts (a comprehension filter; new 2026-10-12)

## Cut



*Cut —; niladic: on error, erases the error and returns Nil*

U+22EE

## Debugger



✔ ⋄ *enters debugger iff error registers set*

U+2714

## Errdmd



*Last error: diamond position in line*

U+2727

## Succeed



*Succeed — like Cut, except returns Truth (new 2026-10-15)*

U+2999

# Anticut



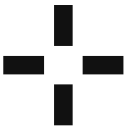
*Anticut — on error throws to debugger; on return from debugger, returns Nil*

U+22EF

# Errors: Monadic

## Silence

141



U+22B9

## Compress

142



U+22D6

## Cut



*Cut —; monadic ⁝ X: checks X for Truth before returning*

U+22EE



# Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Monadic ⊹ V or ⊹ M means Z is identity: keep zeros (or zero-needle signals) as-is, drop nonzeros.



# Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Monadic ⋖ V or ⋖ M means P is identity: pure filter, no transform.

# Classes and Properties: Niladic

## Constant

146

α

U+03B1

## Epsilon

ε

*Epsilon — holds the result of any getter, setter, or nabla when no explicit return glyph is used*

U+03B5

## Omega

ω

*Omega — the vector of arguments passed to a setter (◅)*

U+03C9

# α Constant

U+03B1

Constant — declares a class constant: X α name ← value names a constant for class X. α is also the original (pre-override) value: read-only, may not be changed directly; trying registers an error. As the original value α is an ad hoc constant — contextual to the setter, not a declared class constant

# Classes and Properties: Monadic

## Increment



*Increment —; monadic ⊕ D: invokes D's setter with nabla + 1 as the xthrement.*

U+2295

## Setter



*Setter — the setter nabla: defines a property setter, as ∇ defines a function. Generalized: takes an xthrement argument — modifies instance data (increment/decrement generalized) rather than assigning (new 2026-10-14)*

U+22B2

## Getter



*Getter — the getter nabla: defines a property getter, as ∇ defines a function*

U+22B3

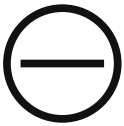
## Property



*Property —; Monadic ⋘ D: yields the result of D's getter (getter mode of D's class call) (new 2026-10-14)*

U+22D8

## Decrement



*Decrement —; monadic ⊖ D: invokes D's setter with nabla - 1 as the xthrement.*

U+2296

## Divide By



*Divide By —; monadic ⊘ D: invokes D's setter with nabla ÷ 2 as the xthrement (halve).*

U+2298

## Multiply By



*Multiply By —; monadic ⊛ D: invokes D's setter with nabla × 2 as the xthrement (double).*

U+229B

# Classes and Properties: Dyadic

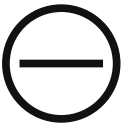
## Increment



*Increment —; Dyadic ⊕ D X: invokes D's setter with nabla + X as the xthrement (new 2026-10-14)*

U+2295

## Decrement



*Decrement —; Dyadic ⊖ D X: invokes D's setter with nabla - X as the xthrement (new 2026-10-14)*

U+2296

## Multiply By



*Multiply By —; Dyadic ⊛ D X: invokes D's setter with nabla ⌿ X as the xthrement (new 2026-10-14)*

U+229B

## Property



*Property —; dyadic ⋘ I D: invokes D's getter (getter mode of D's class call), then invokes I's setter (setter mode of I's class call) with the result as argument; yields the assigned data as a pass-through.*

U+22D8

## Divide By



*Divide By —; Dyadic ⊘ D X: invokes D's setter with nabla ÷ X as the xthrement (new 2026-10-14)*

U+2298

## (method binding)



*Binds method to object: X⊡function*

U+22A1

## Gets



*Gets — ⇔A G: A stored in symbols[G], G stored in instances[A]; bidirectional binding (new 2026-10-13)*

U+2B04

# The Virtual: Niladic

## IntegerBounded



*IntegerBounded — bounded integer class: min/max enforced on creation, assignment, math; clamp + silent sticky flag on violation (new 2026-10-15)*

U+012A

## Trapdoor



*Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).*

U+1D21D

## VirtualSpace



*VirtualSpace — the VR coordinate space: navel origin, forward half-sphere, [r;θ;φ], angles ±32767, distance in mm (new 2026-10-15)*

U+1D4E5

## Prim



*Prim — base VR primitive: name, position, collision machinery; posted via Virtual (new 2026-10-16)*

U+1D7D8

Raster



*Raster — VR prim ⊑ Prim: array of pels, rank 4 [Y,X,1,3] (new 2026-10-16)*

U+1D7D9

Point



*Point — VR prim ⊑ Prim: dot prim, [N,6] (H,V,D,RGB) (new 2026-10-16)*

U+1D7DA

Triangle



*Triangle — VR prim ⊑ Prim: 9 points × 2 thetas + D + RGB = 22 numbers, [N,22] (new 2026-10-16)*

U+1D7DB

# The Virtual: Monadic

Trapdoor

159



U+1D21D

# ⊂ Trapdoor

`U+1D21D`

Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).; Monadic ⊂ A: puts the address of A into the accumulator before calling. ⇔ X ⊂ ∇ assembly ∇: X gets the address of the compiled assembly script. If X is so bound, niladic X calls the code. ε maps to the accumulator. The call yields ε

# Dungeon

# Dungeon: Places

## Where

@

*Where the hero is — bound to '@', the roguelike player glyph. Where⊡Zee yields Where [-1 (last); Where⊡Wye yields Where [1 (midposition — middle of 3 coords); Where⊡Exx yields Where [0 (zero). INDICES START AT ZERO*

U+0040

## Dungeon

£

*The Dungeon — literally the set of all levels*

U+00A3

## Screen

§

*The Screen — intrinsic 20×60 array (16 play + 4 I/O rows)*

U+00A7

## Level

¶

*Level — the pilcrow: a single dungeon level, member of £ Dungeon*

U+00B6

## Cell

¢

*A dungeon cell — the cent sign joins the family: § £ ¢. Cell⊡location ← £ Where: the cell's location is the Dungeon at Where. A dungeon square IS a dictionary of instances (the ⓘ registry's entries present at that square)*

U+00A2

## Up



*Up — the upward/north movement. Where⊡up yields ∅ when Where⊡Zee is 0, else Where with Zee decremented (refined 2026-10-10: boundary-checked — top edge is a quiet Nil, not a negative index)*

U+25B2

## Down



*Down — the downward/south movement, counterpart to Up. Where⊡down yields ∅ when Where⊡Zee ≥ Dungeon⊡Zee (bottom edge), else Where⊡Zee + 1 (confirmed 2026-10-10)*

U+25BC

# Dungeon: Classes

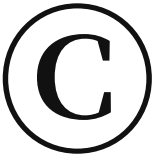
## Class



*Class — the circled dot. X⊡Class gets a RANK-2 ARRAY OF TRAINS: a class IS a decision table made of trains. Class instances are callable; the train defaults to Nil (∅)*

U+2299

## Classes



*Classes — the set of all classes (plural of Class), parallel to £ Dungeon as the set of all levels. Implementation: a single twin (bidirectional word↔glyph) dictionary at global scope*

U+24B8

## Instances

167



U+24BE

## Instance



*Instance — the INDIVIDUAL instance (singular), distinct from Ⓘ the table. Joins the blackboard family (𝕏 𝕐 ℤ ℙ 𝕆). Carries the (glyph, hash-at-creation) identity tuple*

U+1D540



# Instances

U+24BE

Instances — the TABLE/set of all instances (plural), parallel to Ⓒ Classes. Implementation: a single twin (bidirectional word ↔ glyph) dictionary at global scope. Instance identity is the (glyph, hash-at-creation) tuple, used as the key field in both the global instance dictionary and each square (Cell) dictionary

## Interpreter



*Interpreter — the Wuxia interpreter (new 2026-10-15)*

U+203D

## Inherits



*Inherits — the inheritance operator (square subset of)*

U+2291

## Isa



*Isa — the is-a / instantiation operator (2026-10-10). Monadic ∈ Y: Y is a class to copy data from; new instance.ε = Y.ε. Dyadic X ∈ Y X: X is a class, optional initializer; new instance.ε = X.ε (new 2026-10-14)*

U+22FF

## Object



*Object — a DUNGEON object, not a generic OOP object. The blackboard O joins the projection family (𝕏 𝕐 ℤ ℙ): the root of the dungeon's object hierarchy, what every dungeon entity inherits from*

U+1D546

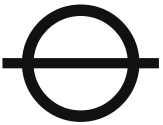
## VirtualDistance



*VirtualDistance — distance in VirtualSpace, 0-512 (new 2026-10-15)*

U+27F0

## Angle



*Angle — a VR angle (theta/phi) (new 2026-10-15)*

U+29B5

## VirtualHorizontal



*VirtualHorizontal — horizontal angle in VirtualSpace (new 2026-10-15)*

U+29D3

## VirtualVertical



*VirtualVertical — vertical angle in VirtualSpace (new 2026-10-15)*

U+29D7

# Dungeon: Objects

## Ring



*Ring — the blackboard R: a dungeon object (ring), third of the dungeon entity glyphs*

U+211D

## Wand



*Wand — left tack: a dungeon object (wand)*

U+22A3

## Armor



*Armor — the blackboard A: a dungeon object (armor), first of the dungeon entity glyphs*

U+1D538

## Weapon



*Weapon — the blackboard W: a dungeon object (weapon), second of the dungeon entity glyphs*

U+1D54E

## Amulet



*Amulet — eight-petalled floret: a dungeon object (amulet). (Was U+1DF76, tofu on device — swapped 2026-10-10)*

U+2741

## Food



*Food — the blackboard F: a dungeon object (food)*

U+1D53D

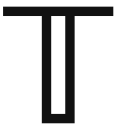
## Scroll



*Scroll — the blackboard S: a dungeon object (scroll)*

U+1D54A

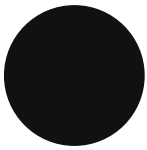
## Tool



*Tool — the blackboard T: a dungeon object (tool)*

U+1D54B

## Gem



*Gem — solid black circle: a dungeon object (gem). (Was U+2733 — swapped 2026-10-10)*

U+25CF

## Potion



*Potion — heavy exclamation: a dungeon object (potion)*

U+2757

# Dungeon: Entities

## Player



*Player — black smiling face: the dungeon's player entity. Inherits Monster plus a few extra fields Monster does not have*

U+263B

## Monster



*Monster — the blackboard M: a dungeon monster, first of the monster glyphs*

U+1D544

# Dungeon: Food

## Tin



*Tin — circled white star: a dungeon food item (tin), first of the object species*

U+272A

## RefriedBeans



*RefriedBeans — dark-encircled six: grants teleportation*

U+277B

## Spinach



*Spinach — dark-encircled seven: Popeye strength*

U+277C

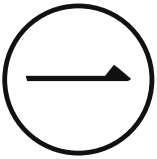
## Sushi



*Sushi — circled congratulation: the sushi/sashimi food category. (Was U+2722 — swapped 2026-10-11)*

U+3297

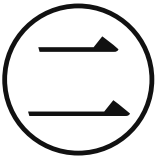
## CarpSashimi



*CarpSashimi — circled kanji: sushi item*

U+3280

## WhaleSashimi



*WhaleSashimi — circled kanji: sushi item*

U+3281

## EelSashimi



*EelSashimi — circled kanji: sushi item*

U+3282

## SalmonSashimi



*SalmonSashimi — circled kanji: sushi item*

U+3283

## TunaSashimi



*TunaSashimi — circled kanji: sushi item*

U+3284

## CrabSashimi



*CrabSashimi — circled kanji: sushi item*

U+3285

## DragonSashimi



*DragonSashimi — circled kanji: sushi item*

U+3286

## TurtleSashimi



*TurtleSashimi — circled kanji: sushi item*

U+3287

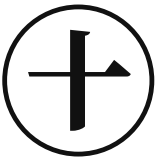
## Oyster



*Oyster — circled kanji: sushi item*

U+3288

## Uni



*Uni — circled kanji: sushi item*

U+3289

## Gyuniku



*Gyuniku — circled kanji: sushi item*

U+328A

## Asian



*Asian — squared hand: the Asian food category. (Was U+328C — swapped 2026-10-11)*

U+1F210

Ramen



*Ramen — squared cjk unified ideograph-5b57: Asian food*

U+1F211

FortuneCookie



*FortuneCookie — squared cjk unified ideograph-53cc: Asian food*

U+1F212

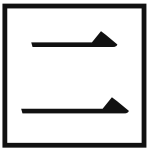
Kelp



*Kelp — squared katakana de: Asian food*

U+1F213

Dumplings



*Dumplings — squared cjk unified ideograph-4e8c: Asian food*

U+1F214

BowlOfRice



*BowlOfRice — squared cjk unified ideograph-591a: Asian food*

U+1F215

SpringRolls



*SpringRolls — squared cjk unified ideograph-89e3: Asian food*

U+1F216

Tofu



*Tofu — squared cjk unified ideograph-5929: Asian food*

U+1F217

PreservedEgg



*PreservedEgg — squared cjk unified ideograph-4ea4: Asian food*

U+1F218

## Soup



*Soup — circled twenty: the soup category*

U+2473

## Deli



*Deli — black diamond minus white X: the delicatessen food category (the 25 staples, names lost 2026-10-10 — rebuild pending)*

U+2756

## Fruit



*Fruit — domino tile back: the fruit food category*

U+1F030

## CantoneseDuck




*CantoneseDuck — squared cjk unified ideograph-6620: Asian food*

U+1F219

## Egg

Ø

*Egg — o with stroke: the egg. (Was U+2800 — swapped 2026-10-11)*

U+00F8

## Meat



*Meat — CJK beautiful: the meat category*

U+4E3D

## Creature



*Creature — o with long stroke overlay: the monster-shuffle category*

U+A74B

## Candy




*Candy — mahjong joker: the candy category*

U+1F02A

## Goodbar



*Goodbar — mahjong east wind: candy*

`U+1F000`

## Honee



*Honee — mahjong south wind: candy*

`U+1F001`

## DRation



*DRation — mahjong west wind: candy*

`U+1F002`

## DoveBar



*DoveBar — mahjong north wind: candy*

`U+1F003`

## AcidBroth

❶

*AcidBroth — dark-encircled number: soup*

U+2776

## GoldenSoup

❷

*GoldenSoup — dark-encircled number: soup*

U+2777

## StonedSoup

❸

*StonedSoup — dark-encircled number: soup*

U+2778

## CrowBar




*CrowBar — mahjong winter: candy*

U+1F029




## MoldySoup

*MoldySoup — dark-encircled number: soup*

U+2779




## Manwich

*Manwich — dark-encircled number: soup*

U+277A




## CreamOfAlgaeSoup

*CreamOfAlgaeSoup — dark-encircled number: soup*

U+277E




## ChickenNoodleSoup

*ChickenNoodleSoup — dark-encircled number: soup*

U+277F

## Apple



*Apple — domino tile: fruit, inherits Fruit*

U+1F032

## Orange



*Orange — domino tile: fruit, inherits Fruit*

U+1F033

## Pear



*Pear — domino tile: fruit, inherits Fruit*

U+1F034

## Melon



*Melon — domino tile: fruit, inherits Fruit*

U+1F035

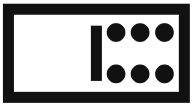
## Banana



*Banana — domino tile: fruit, inherits Fruit*

U+1F036

## Carrot



*Carrot — domino tile: fruit, inherits Fruit*

U+1F037

## Fig



*Fig — domino tile: fruit, inherits Fruit*

U+1F038

## Lemon



*Lemon — domino tile: fruit, inherits Fruit*

U+1F039

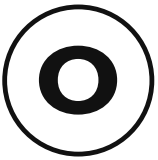
## Tomb




*Tomb — circled 10 on black square: safe-row tin, now Sundry*

U+3248

## Sundry



*Sundry — circled ideograph: the sundry/miscellaneous category*

U+3267

## Lime



*Lime — domino tile: fruit, inherits Fruit*

U+1F03A

## Grapes



*Grapes — domino tile: fruit, inherits Fruit*

U+1F03B

## Pesto




*Pesto — circled 20 on black square: safe-row tin, now Sundry*

U+3249

## KelpFrond




*KelpFrond — circled water: sundry item (kelp is seaweed)*

U+328C

## EucalyptusLeaf




*EucalyptusLeaf — circled wood: sundry item*

U+328D

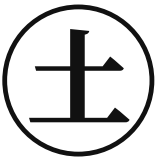
## CandyBar




*CandyBar — circled metal: sundry item*

U+328E

LembasWafer




*LembasWafer — circled earth: sundry item*

U+328F

CramRation




*CramRation — circled sun: sundry item*

U+3290

FoodRation




*FoodRation — circled medicine: sundry item*

U+32A9

KRation




*KRation — circled religion: sundry item*

U+32AA

## CloveOfGarlic




*CloveOfGarlic — circled attention: sundry item*

U+329F

## SprigOfWolfsbane




*SprigOfWolfsbane — circled item: sundry item*

U+32A0

## LumpOfRoyalJelly




*LumpOfRoyalJelly — circled copy: sundry item*

U+32A2

## CRation




*CRation — circled study: sundry item*

U+32AB

# Dungeon: Species inheritance

## EarthElemental



*⊑ Elementals — Species inherits its category (2026-10-10)*

U+2295

## FireElemental



*⊑ Elementals — Species inherits its category (2026-10-10)*

U+229B

## AirElemental



*⊑ Elementals — Species inherits its category (2026-10-10)*

U+2A37

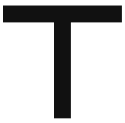
## WaterElemental



*⊑ Elementals — Species inherits its category (2026-10-10)*

U+2A38

## PurpleWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+22A4

## BabyPurpleWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+22A6

## LongWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+2A53

## BabyLongWorm



*⊑ Worms — Species inherits its category (2026-10-10)*

U+2A55

## LargeKobold



*⊑ Kobolds — Species inherits its category (2026-10-10)*

U+049C

## Kobold



*⊑ Kobolds — Species inherits its category (2026-10-10)*

U+049D

## MasterMindFlayer



*⊑ Mindflayers — Species inherits its category (2026-10-10)*

U+04F4

## MindFlayer



*⊑ Mindflayers — Species inherits its category (2026-10-10)*

U+04F5

## StoneGiant

Ġ

*⊑ Giants — Species inherits its category (2026-10-10)*

U+0120

## Giant

Ģ

*⊑ Giants — Species inherits its category (2026-10-10)*

U+0122

## KoboldLord

Ќ

*⊑ Kobolds — Species inherits its category (2026-10-10)*

U+040C

## KoboldShaman

Ҝ

*⊑ Kobolds — Species inherits its category (2026-10-10)*

U+049B

## FireGiant

Ĝ

*⊑ Giants — Species inherits its category (2026-10-10)*

`U+011C`

## HillGiant

Ğ

*⊑ Giants — Species inherits its category (2026-10-10)*

`U+011E`

## FrostGiant

Ǵ

*⊑ Giants — Species inherits its category (2026-10-10)*

`U+0193`

## Ettin

Gˇ

*⊑ Giants — Species inherits its category (2026-10-10)*

`U+01E6`

## VampireBat

ƀ

*⊑ Vampires — Species inherits its category (2026-10-10)*

U+0180

## StormGiant

Ǧ

*⊑ Giants — Species inherits its category (2026-10-10)*

U+01E4

## Vampire

Ỿ

*⊑ Vampires — Species inherits its category (2026-10-10)*

U+1E7F

## Titan

⅁

*⊑ Giants — Species inherits its category (2026-10-10)*

U+2141

## VampireLeader



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+1E7D

## VampireLady



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

## VampireLord



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

## Nosferatu



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

## HumanMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## ElfMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## DwarfMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## GnomeMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## OrcMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## KoboldMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## GiantMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## EttinMummy



*⊑ Undead — Species inherits its category (2026-10-10)*

U+046A

## KoboldZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## GnomeZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## OrcZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## DwarfZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## ElfZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## HumanZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## GiantZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## EttinZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

## Ghoul

*g*

*⊑ Undead — Species inherits its category (2026-10-10)*

U+210A

## WoodElf



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2618

## GreenElf



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2618

## GreyElf



*⊑ Elves — Species inherits its category (2026-10-10)*

U+2618

## BabyDragon

⁇

*⊏ Dragons — Species inherits its category (2026-10-10)*

U+2047

## Elf



*⊏ Elves — Species inherits its category (2026-10-10)*

U+2618

## ElfNoble



*⊏ Elves — Species inherits its category (2026-10-10)*

U+2E19

## ElvenMonarch

支

*⊏ Elves — Species inherits its category (2026-10-10)*

U+2F40

## Centipede

¢

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00A2

## CaveSpider

¤

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00A4

## Scorpion

±

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00B1

## Dragon

⁇

*⊑ Dragons — Species inherits its category (2026-10-10)*

U+2047

## GiantSpider

Ø

*⊑ Spiders — Species inherits its category (2026-10-10)*

U+00D8

## Mastodon

Ψ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+03A8

## Mumak

Ψ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+03A8

## Baluchitherium

ϒ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+03D2

## Wumpus

¿

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+00BF

## Trapper



*⊑ Beasts — Species inherits its category (2026-10-10)*

U+048A

## Minotaur

Ṁ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+1E40

## Titanothere



*⊑ Beasts — Species inherits its category (2026-10-10)*

U+213A

## Bugbear

ö

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+00F6

## Owlbear

ø

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+00F8

## Jabberwock

ƹ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+01B8

## Rothe

ȣ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+0222

## Shade

@

*⊑ Guises — Species inherits its category (2026-10-10)*

U+0040

## Leocrotta

ꝉ

*⊑ Beasts — Species inherits its category (2026-10-10)*

U+023D

## Stalker

Ǝ

*⊑ Lurkers — Species inherits its category (2026-10-10)*

U+042D

## LurkerAbove



*⊑ Lurkers — Species inherits its category (2026-10-10)*

U+214F

## Skeleton



*⊑ Guises — Species inherits its category (2026-10-10)*

U+0040

## Foocubus



*⊑ Guises — Species inherits its category (2026-10-10)*

U+0040

## MountainNymph



*⊑ Guises — Species inherits its category (2026-10-10)*

U+0040

## WoodNymph



*⊑ Guises — Species inherits its category (2026-10-10)*

U+0040

## WaterNymph

@

*⊑ Guises — Species inherits its category (2026-10-10)*

U+0040

## MerfolkRoyal

⁇

*⊑ Merfolk — Species inherits its category (2026-10-10)*

U+2047

## MerfolkNoble

⁇

*⊑ Merfolk — Species inherits its category (2026-10-10)*

U+2047

## MerfolkWitch

⁇

*⊑ Merfolk — Species inherits its category (2026-10-10)*

U+2047

## WizardOfYendor

¥

*⊑ Specials — Species inherits its category (2026-10-10)*

U+00A5

## Shopkeeper

§

*⊑ Specials — Species inherits its category (2026-10-10)*

U+00A7

## MerfolkCommon

⁇

*⊑ Merfolk — Species inherits its category (2026-10-10)*

U+2047

## GrandMaster

⁇

*⊑ Specials — Species inherits its category (2026-10-10)*

U+2047

## Tin



*⊏ Food — Tin inherits Food — first object species joins the hierarchy (2026-10-10)*

U+272A

## AcidBroth



*⊏ Soup — Soup inherits Soup (2026-10-10)*

U+2776

## GoldenSoup



*⊏ Soup — Soup inherits Soup (2026-10-10)*

U+2777

## StonedSoup



*⊏ Soup — Soup inherits Soup (2026-10-10)*

U+2778

4

## MoldySoup

*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+2779

5

## Manwich

*⊑ Soup — Soup inherits Soup — the human-flesh tin (2026-10-10)*

U+277A

9

## CreamOfAlgaeSoup

*⊑ Soup — Soup inherits Soup (2026-10-10)*

U+277E

10

## ChickenNoodleSoup

*⊑ Soup — Soup inherits Soup (2026-10-10)*

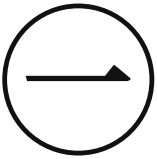
U+277F



## RefriedBeans

*⊑ Soup — Soup item inherits Soup — grants teleportation (2026-10-11)*

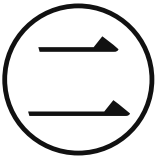
U+277B



## CarpSashimi

*⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3280



## WhaleSashimi

*⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3281



## Sushi

*⊑ Food — Sushi inherits Food — the sushi food category (2026-10-10)*

U+3297

## EelSashimi



*㆓ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3282

## SalmonSashimi



*㆓ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3283

## TunaSashimi



*㆓ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3284

## CrabSashimi



*㆓ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3285

## DragonSashimi



*⿲ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3286

## TurtleSashimi



*⿲ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3287

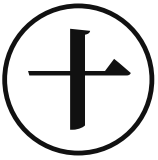
## Oyster



*⿲ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3288

## Uni



*⿲ Sushi — Individual sushi*
*inherits Sushi (2026-10-10)*

U+3289

## Gyuniku



*⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+328A

## Asian



*⊑ Food — Asian inherits Food — the Asian food category (2026-10-10)*

U+1F210

## Ramen



*⊑ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F211

## FortuneCookie



*⊑ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F212

## Kelp



*⿺ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F213

## Dumplings



*⿺ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F214

## BowlOfRice



*⿺ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F215

## SpringRolls



*⿺ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F216

## Deli



*⿸ Food — Deli inherits Food — the delicatessen food category (2026-10-10)*

U+2756

## Tofu



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F217

## PreservedEgg



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F218

## CantoneseDuck



*⿸ Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F219

## KelpFrond



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328C

## EucalyptusLeaf



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328D

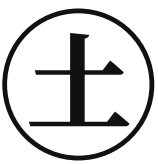
## CandyBar



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328E

## LembasWafer



*⿸ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328F

㊐

CramRation

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+3290

㊐

FoodRation

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A9

㊐

KRation

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AA

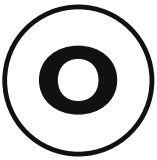
㊐

CRation

*㊐ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AB

## Sundry



*㆓ Deli — Sundry is part of Deli (2026-10-10)*

U+3267

## CloveOfGarlic



*㆓ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+329F

## SprigOfWolfsbane



*㆓ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A0

## LumpOfRoyalJelly



*㆓ Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A2

## Fruit



*⊑ Food — Fruit inherits Food — the fruit food category (2026-10-10)*

U+1F030

## Apple



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F032

## Orange



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F033

## Pear



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F034

## Melon



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F035

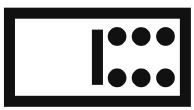
## Banana



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F036

## Carrot



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F037

## Fig



*⊑ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F038

## Soup



*⊏ Tin — Soup inherits Tin — the soup category (2026-10-11)*

U+2473

## Lemon



*⊏ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F039

## Lime



*⊏ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03A

## Grapes



*⊏ Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03B

## Goodbar



*⊑ Candy — Goodbar inherits Candy (2026-10-11)*

U+1F000

## Honee



*⊑ Candy — Honee inherits Candy (2026-10-11)*

U+1F001

## DRation



*⊑ Candy — DRation inherits Candy (2026-10-11)*

U+1F002

## Candy



*⊑ Food — Candy inherits Food — the candy category (2026-10-11)*

U+1F02A

## Tomb



*⿺ Sundry — Tomb inherits Sundry*
*(2026-10-11)*

U+3248

## Pesto



*⿺ Sundry — Pesto inherits Sundry*
*(2026-10-11)*

U+3249

## DoveBar



*⿺ Candy — DoveBar inherits Candy*
*(2026-10-11)*

U+1F003

## CrowBar



*⿺ Candy — CrowBar inherits Candy*
*(2026-10-11)*

U+1F029

## Egg

Ø

*⊑ Creature — Egg inherits Creature (2026-10-11)*

U+00F8

## Spinach



*⊑ Soup — Spinach (NetHack tin filling, Popeye strength) inherits Soup (2026-10-11)*

U+277C

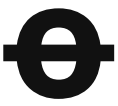
## Meat



*⊑ Creature — Meat inherits Creature — the meat category (2026-10-11)*

U+4E3D

## Creature



*⊑ Tin — Creature inherits Tin — the creature-meat tin category; the category that shuffles monsters as consumables (tins of monster meat) (2026-10-11)*

U+A74B

# IntegerBounded

Ī

⊑ *Integer — IntegerBounded inherits Integer (new 2026-10-15)*

U+012A

# IntegerBounded.max

⁇

⊑ *Integer — IntegerBounded.max is an Integer (new 2026-10-15)*

U+2047

# IntegerBounded.min

⁇

⊑ *Integer — IntegerBounded.min is an Integer (new 2026-10-15)*

U+2047

# IntegerBounded.epsilon

⁇

⊑ *Integer — IntegerBounded.epsilon is an Integer (new 2026-10-15)*

U+2047

## IntegerBounded.nabla

⁇

*IntegerBounded.nabla enforces bounded gets and sets (new 2026-10-15)*

U+2047

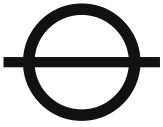
## IntegerBounded.nabla

⁇

*⊑ Interpreter — IntegerBounded.nabla is an Interpreter (new 2026-10-15)*

U+2047

## Angle



*⊑ IntegerBounded — Angle inherits IntegerBounded (new 2026-10-15)*

U+29B5

## VirtualHorizontal



*⊑ Angle — VirtualHorizontal inherits Angle (new 2026-10-15)*

U+29D3

## Egg

Ø

*Egg stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)*

U+00F8

## VirtualDistance



*⊑ IntegerBounded — VirtualDistance inherits IntegerBounded (new 2026-10-15)*

U+27F0

## VirtualVertical



*⊑ Angle — VirtualVertical inherits Angle (new 2026-10-15)*

U+29D7

## Meat



*Meat stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)*

U+4E3D

# Index

# English Index


AcidBroth 189-1
AcidBroth 220-2
AirElemental 199-3
Amulet 173-1
Anagram 75-4
Anagram 82-4
Angle 169-2
Angle 238-3
Anticut 138-1
Antilog 28-2
Apple 191-1
Apple 231-2
Applyself 122-1
Approx 13-3
Arccosine 26-1
Arcsine 26-2
Arctangent 26-4
Armor 172-3
Asian 183-4
Asian 225-2
Atomize 110-2
BabyDragon 211-1
BabyLongWorm 200-4
BabyPurpleWorm 200-2
Backstroke 108-1
Backstroke 113-1
Backstroke (forward) 108-3
Backstroke (forward) 113-3
Baluchitherium 213-4
Banana 192-1
Banana 232-2
Base Ten Log 29-4
Bell 39-3
Bell 44-3
Bell 47-1
BowlOfRice 185-1
BowlOfRice 226-3
Box 87-1
Box 89-1
Box 90-1
Box 95-1
Bucket 103-2
Bugbear 215-1
Caboose 110-4
Candy 187-4
Candy 234-4
CandyBar 194-4
CandyBar 228-3
CantoneseDuck 186-4
CantoneseDuck 227-4
Cardinality 80-1
CarpSashimi 181-1
CarpSashimi 222-2
Carrot 192-2
Carrot 232-3
Catalogue 89-3
Catalogue 93-3
Catenate 73-2
Catenate 79-3
Catenate 83-1
CaveSpider 212-2
Ceiling 25-1
Ceiling 35-2
Cell 163-1
Center 80-2
Centipede 212-1
ChickenNoodleSoup 190-4
ChickenNoodleSoup 221-4
Class 166-1
Classes 166-2
CloveOfGarlic 196-1
CloveOfGarlic 230-2
Collate 130-1
Combinatoric 51-3
Combinatoric 54-1
Compress 134-2

Compress 140-2
Congruent 9-3
Congruent 11-3
Conjugate 21-3
Constant 145-1
Cosine 25-4
Count 76-3
CrabSashimi 182-2
CrabSashimi 223-4
CramRation 195-2
CramRation 229-1
CRation 196-4
CRation 229-4
CreamOfAlgaeSoup 190-3
CreamOfAlgaeSoup 221-3
Creature 187-3
Creature 236-4
CrowBar 189-4
CrowBar 235-4
Curry 130-2
Cut 137-1
Cut 140-3
D2 41-2
D4 41-3
D6 41-4
Deal 45-1
Debugger 137-2
Decide 125-3
Decrement 149-1
Decrement 151-2
Deli 186-2
Deli 227-1
Determinant 67-1
Dice 40-4
Dice 44-4
Dice 47-2
Diction 125-1
Differentiate 57-3
Divide (Complex) 18-3
Divide (Complex) 23-3
Divide (Complex) 33-3
Divide (Polar) 18-4
Divide (Polar) 23-4
Divide (Polar) 33-4
Divide By 149-2
Divide By 152-1
Divisible 50-1
Divisible 53-2
Double 21-4
DoveBar 188-4
DoveBar 235-3
Down 163-3
Dragon 212-4
DragonSashimi 182-3
DragonSashimi 224-1
DRation 188-3
DRation 234-3
Drop 83-3
Drop Train 120-3
Dumplings 184-4
Dumplings 226-2
Dungeon 162-2
Duplicate 110-3
Duplicate Head 113-2
Duplicate Tail 115-3
DwarfMummy 206-3
DwarfZombie 208-4
EarthElemental 199-1
EelSashimi 181-3
EelSashimi 223-1
Egg 187-1
Egg 236-1
Egg 239-1
Elf 211-2
ElfMummy 206-2
ElfNoble 211-3
ElfZombie 209-1
ElvenMonarch 211-4
Enlist 101-1
Enumerate 76-1
Epsilon 145-2
Equality 12-1
Errdmd 137-3
Errline 134-3

Errmsg 134-4
Ettin 203-4
EttinMummy 207-4
EttinZombie 209-4
EucalyptusLeaf 194-3
EucalyptusLeaf 228-2
Euler 40-3
Evaluate 60-4
Extend 73-1
Extend 79-2
Extend 84-2
Exx 72-3
Factor 51-2
Factorial 19-1
Factorial 29-1
Factorial 35-1
Fetch 93-2
Fig 192-3
Fig 232-4
FireElemental 199-2
FireGiant 203-1
FixedPoint 61-1
Floor 25-2
Floor 35-3
Foocubus 217-2
Food 173-2
FoodRation 195-3
FoodRation 229-2
forall 13-2
Fork 120-2
Fornone 14-2
Forsome 14-1
FortuneCookie 184-2
FortuneCookie 225-4
Fourier 29-2
FrostGiant 203-3
Fruit 186-3
Fruit 231-1
Function 125-4
Gamma 40-1
GCD 53-3
Gem 174-1
Gets 152-3
Getter 148-3
Ghoul 210-1
Giant 202-2
GiantMummy 207-3
GiantSpider 213-1
GiantZombie 209-3
Glyph 125-2
GnomeMummy 206-4
GnomeZombie 208-2
Golden 40-2
GoldenSoup 189-2
GoldenSoup 220-3
Goodbar 188-1
Goodbar 234-1
Grade Down 73-4
Grade Up 73-3
GrandMaster 219-4
Grapes 193-4
Grapes 233-4
Greater Or Equal 12-4
Greater Than 11-2
GreenElf 210-3
GreyElf 210-4
Gyuniku 183-3
Gyuniku 225-1
Half 39-1
Half 44-1
Head 115-1
HillGiant 203-2
Honee 188-2
Honee 234-2
HumanMummy 206-1
HumanZombie 209-2
Hyperbolic Arccosine 28-4
Hyperbolic Arcsine 27-3
Hyperbolic Arctangent 28-3
Hyperbolic Cosine 27-2
Hyperbolic Sine 27-1
Hyperbolic Tangent 27-4
Identity 107-4
Implies 13-1

In 103-1
In 104-1
InAt 103-4
Increment 148-1
Increment 151-1
Indivisible 50-2
Indivisible 53-4
Inherits 168-2
InLast 103-3
Inner Product 64-2
Inner Product 66-2
Inner Product 69-2
InOf 100-3
Instance 166-4
Instances 166-3
Integer Division 60-3
IntegerBounded 155-1
IntegerBounded 237-1
IntegerBounded.epsilon 237-4
IntegerBounded.max 237-2
IntegerBounded.min 237-3
IntegerBounded.nabla 238-1
IntegerBounded.nabla 238-2
Integrate 57-4
Interpreter 168-1
Intersection 98-1
Intersection 100-1
Intersection 104-2
InverseFourier 29-3
Invert 9-1
Isa 168-3
IsAt 11-4
Jabberwock 215-3
Kelp 184-3
Kelp 226-1
KelpFrond 194-2
KelpFrond 228-1
Kobold 201-2
KoboldLord 202-3
KoboldMummy 207-2
KoboldShaman 202-4
KoboldZombie 208-1
Kount 76-4
KRation 195-4
KRation 229-3
L Factor 66-4
Lambda 128-1
Laminate 75-3
Laminate 82-3
LargeKobold 201-1
Lay 127-4
LCM 53-1
LembasWafer 195-1
LembasWafer 228-4
Lemon 192-4
Lemon 233-2
Leocrotta 216-2
Less Or Equal 12-3
Less Than 11-1
Level 162-4
Lime 193-3
Lime 233-3
Link 93-4
(literalize) 127-2
Local 58-3
LongWorm 200-3
LumpOfRoyalJelly 196-3
LumpOfRoyalJelly 230-4
LurkerAbove 216-4
Magnitude 30-1
Manwich 190-2
Manwich 221-2
Map 107-2
Map 112-2
MasterMindFlayer 201-3
Mastodon 213-2
Meat 187-2
Meat 236-3
Meat 239-4
Melon 191-4
Melon 232-1
MerfolkCommon 219-3
MerfolkNoble 218-3
MerfolkRoyal 218-2

MerfolkWitch 218-4
Meter 110-1
(method binding) 152-2
MindFlayer 201-4
Minotaur 214-3
Minus (Complex) 18-1
Minus (Complex) 23-1
Minus (Complex) 33-1
MoldySoup 190-1
MoldySoup 221-1
Monster 177-2
MountainNymph 217-3
Multiply 19-4
Multiply 24-4
Multiply 34-4
Multiply By 149-3
Multiply By 151-3
Multiply Matrices 64-3
Multiply Matrices 66-3
Multiply Matrices 69-3
Mumak 213-3
Natural Log 28-1
Natural Log 35-4
Negate 21-2
Negate / Subtract 17-2
NextLocal 57-1
Nil 7-2
Nip 119-1
Nosferatu 205-4
Not 9-2
Object 168-4
Omega 145-3
Optimize 58-2
Optimize 60-2
Orange 191-2
Orange 231-3
OrcMummy 207-1
OrcZombie 208-3
Outer Product 64-1
Outer Product 66-1
Outer Product 69-1
Over 119-4
Owlbear 215-2
Oyster 183-1
Oyster 224-3
Pear 191-3
Pear 231-4
Pee 72-1
Percent 19-2
Percent 24-1
Percent 34-1
Permute 75-2
Permute 82-2
Pesto 194-1
Pesto 235-2
Pi 39-2
Pick 42-1
Pick 44-2
Pipeline 113-4
Pipeline 122-2
Player 177-1
Plus 32-1
Plus (Complex) 17-3
Plus (Complex) 22-3
Plus (Complex) 32-3
Point 156-2
Potion 174-2
PreservedEgg 185-4
PreservedEgg 227-3
Prim 155-4
Prime 50-4
Prime 51-1
Primes 50-3
Print 127-1
Property 148-4
Property 151-4
PurpleWorm 200-1
Put 82-1
Quotient 24-3
Quotient 34-3
Rail 127-3
Ramen 184-1
Ramen 225-3
Rank 75-1

Raster 156-1
Ravel 78-2
RavelRows 78-1
Raze 89-4
Real 39-4
Reduce 107-1
Reduce 112-1
Reduce (forward) 107-3
Reduce (forward) 112-3
Reflect 76-2
RefriedBeans 180-2
RefriedBeans 222-1
Replicate 117-2
Reveal 60-1
Reverse 78-3
Reverse 83-4
Reverse Train 119-3
Ring 172-1
Roll Percentile 41-1
Roots 57-2
Rotate 79-4
Rotate 84-3
Rotate Head 117-3
Rotate Tail 117-4
Rothe 215-4
SalmonSashimi 181-4
SalmonSashimi 223-2
Scorpion 212-3
Screen 162-3
Scroll 173-3
Set Difference 104-4
Setter 148-2
Shade 216-1
Shape 79-1
Shape 84-1
Shopkeeper 219-2
Sight 80-3
Signum 21-1
Signum / Plus 17-1
Silence 134-1
Silence 140-1
Sine 25-3
Skeleton 217-1
Solve 58-1
Soup 186-1
Soup 233-1
Spinach 180-3
Spinach 236-2
SprigOfWolfsbane 196-2
SprigOfWolfsbane 230-3
SpringRolls 185-2
SpringRolls 226-4
Square 19-3
Square 36-1
Square Root 22-2
Stalker 216-3
Star 24-2
Star 34-2
StonedSoup 189-3
StonedSoup 220-4
StoneGiant 202-1
Stop 108-2
StormGiant 204-2
Strip 108-4
Strip 115-4
Subtract 32-2
Succeed 137-4
Sum 22-1
Summarize 87-3
Summarize 90-3
Summarize 93-1
Sundry 193-2
Sundry 230-1
Sushi 180-4
Sushi 222-4
Swap 117-1
Tail 115-2
Take 83-2
Tangent 26-3
Times (Complex) 17-4
Times (Complex) 22-4
Times (Complex) 32-4
Times (Polar) 18-2
Times (Polar) 23-2

Times (Polar) 33-2
Tin 180-1
Tin 220-1
Titan 204-4
Titanothere 214-4
Tofu 185-3
Tofu 227-2
Tomb 193-1
Tomb 235-1
Tool 173-4
Transpose 78-4
Trapdoor 155-2
Trapdoor 158-1
Trapper 214-2
Triangle 156-3
Triple 120-1
Truth 7-3
Tuck 119-2
TunaSashimi 182-1
TunaSashimi 223-3
TurtleSashimi 182-4
TurtleSashimi 224-2
U Factor 67-2
UnApprox 13-4
Unbox 87-2
Unbox 89-2
Unbox 90-2
Unbox 95-2
Unequal 12-2
Uni 183-2
Uni 224-4
Union 98-2
Union 100-2
Union 104-3
Up 163-2
Vampire 204-3
VampireBat 204-1
VampireLady 205-2
VampireLeader 205-1
VampireLord 205-3
VirtualDistance 169-1
VirtualDistance 239-2
VirtualHorizontal 169-3
VirtualHorizontal 238-4
VirtualSpace 155-3
VirtualVertical 169-4
VirtualVertical 239-3
Wand 172-2
WaterElemental 199-4
WaterNymph 218-1
Weapon 172-4
WhaleSashimi 181-2
WhaleSashimi 222-3
Where 100-4
Where 162-1
WizardOfYendor 219-1
WoodElf 210-2
WoodNymph 217-4
Wumpus 214-1
Wuxia 7-1
Wye 72-4
Zee 72-2
Zero 112-4
Zip 90-4

# Glyph Index


! (U+0021) 19-1, 29-1, 35-1
% (U+0025) 19-2, 24-1, 34-1
& (U+0026) 58-1
* (U+002A) 24-2, 34-2
+ (U+002B) 17-1, 21-1, 32-1
- (U+002D) 17-2, 21-2, 32-2
/ (U+002F) 107-1, 112-1
< (U+003C) 11-1
= (U+003D) 12-1
> (U+003E) 11-2
? (U+003F) 45-1
@ (U+0040) 162-1, 216-1, 217-1, 217-2, 217-3 217-4, 218-1
\ (U+005C) 108-1, 113-1
~ (U+007E) 9-1
¡ (U+00A1) 125-1
¢ (U+00A2) 163-1, 212-1
£ (U+00A3) 162-2
¤ (U+00A4) 212-2
¥ (U+00A5) 219-1
§ (U+00A7) 162-3, 219-2
¬ (U+00AC) 9-2
® (U+00AE) 125-2
± (U+00B1) 130-1, 212-3
¶ (U+00B6) 162-4
½ (U+00BD) 39-1, 44-1
¿ (U+00BF) 125-3, 214-1
Ø (U+00D8) 213-1
ö (U+00F6) 215-1
÷ (U+00F7) 24-3, 34-3
ø (U+00F8) 187-1, 215-2, 236-1, 239-1
Ĝ (U+011C) 203-1
ğ (U+011E) 203-2
Ġ (U+0120) 202-1
Ģ (U+0122) 202-2
Ī (U+012A) 155-1, 237-1
ƀ (U+0180) 204-1
Ɠ (U+0193) 203-3
Ƹ (U+01B8) 215-3
Ǥ (U+01E4) 204-2
Ǧ (U+01E6) 203-4
Ȣ (U+0222) 215-4
Ƚ (U+023D) 216-2
Λ (U+039B) 128-1
Ψ (U+03A8) 213-2, 213-3
α (U+03B1) 145-1
γ (U+03B3) 40-1
ε (U+03B5) 145-2
π (U+03C0) 39-2
ρ (U+03C1) 79-1, 84-1
ϕ (U+03C6) 40-2
ω (U+03C9) 145-3
ϒ (U+03D2) 213-4
Ϝ (U+03DC) 29-2
Ќ (U+040C) 202-3
Э (U+042D) 216-3
Ѫ (U+046A) 206-1, 206-2, 206-3, 206-4, 207-1 207-2, 207-3, 207-4
Ѝ (U+048A) 214-2
ҝ (U+049B) 202-4
Ҟ (U+049C) 201-1
ҟ (U+049D) 201-2
Ӄ (U+04F4) 201-3
ӄ (U+04F5) 201-4
Ṁ (U+1E40) 214-3
ṽ (U+1E7D) 205-1
ỿ (U+1E7F) 204-3
‰ (U+2030) 41-1
⁝ (U+203D) 168-1
⁇ (U+2047) 205-2, 205-3, 205-4, 211-1, 212-4 218-2, 218-3, 218-4, 219-3, 219-4 237-2, 237-3, 237-4, 238-1, 238-2
Ɛ (U+2107) 40-3
ℊ (U+210A) 210-1
ℓ (U+2113) 57-1
№ (U+2116) 76-1

| | | |
|---|---|---|
| ℙ | (U+2119) | 72-1 |
| ℜ | (U+211C) | 60-1 |
| ℝ | (U+211D) | 172-1 |
| ℞ | (U+211E) | 57-2 |
| ℤ | (U+2124) | 72-2, 208-1, 208-2, 208-3, 208-4 209-1, 209-2, 209-3, 209-4 |
| ℨ | (U+2128) | 29-3 |
| ℭ | (U+213A) | 214-4 |
| ℾ | (U+213E) | 73-1, 79-2, 84-2 |
| ℿ | (U+213F) | 73-2, 79-3, 83-1 |
| ⅁ | (U+2141) | 204-4 |
| ⅋ | (U+214B) | 58-2, 60-2 |
| ⅏ | (U+214F) | 216-4 |
| Ⅱ | (U+2161) | 113-2 |
| Ⅲ | (U+2162) | 120-1 |
| ↑ | (U+2191) | 83-2 |
| ↓ | (U+2193) | 83-3 |
| ⇋ | (U+21CB) | 117-1 |
| ⇒ | (U+21D2) | 13-1 |
| ↤ | (U+21F7) | 119-1 |
| ↦ | (U+21F8) | 119-2 |
| ↭ | (U+21F9) | 119-3 |
| ∀ | (U+2200) | 13-2 |
| ∂ | (U+2202) | 57-3 |
| ∃ | (U+2203) | 14-1 |
| ∄ | (U+2204) | 14-2 |
| ∅ | (U+2205) | 7-2 |
| ∇ | (U+2207) | 125-4 |
| ∈ | (U+2208) | 103-1 |
| ϵ | (U+220A) | 101-1, 104-1 |
| ∐ | (U+2210) | 80-1, 103-2 |
| ∑ | (U+2211) | 22-1 |
| ∔ | (U+2214) | 110-1 |
| √ | (U+221A) | 22-2 |
| ∩ | (U+2229) | 98-1, 100-1, 104-2 |
| ∪ | (U+222A) | 98-2, 100-2, 104-3 |
| ∫ | (U+222B) | 57-4 |
| ∷ | (U+2237) | 110-2 |
| ÷ | (U+223B) | 76-2 |
| ≈ | (U+2248) | 13-3 |
| ≉ | (U+2249) | 13-4 |
| ≖ | (U+2256) | 89-1 |
| ≗ | (U+2257) | 89-2 |
| ≠ | (U+2260) | 12-2 |
| ≡ | (U+2261) | 9-3, 11-3 |
| ≣ | (U+2263) | 11-4 |
| ≤ | (U+2264) | 12-3 |
| ≥ | (U+2265) | 12-4 |
| ⋬ | (U+226C) | 107-2, 112-2 |
| ⊂ | (U+2282) | 87-1, 90-1, 95-1 |
| ⊃ | (U+2283) | 87-2, 90-2, 95-2 |
| ⊏ | (U+2291) | 168-2 |
| ⊕ | (U+2295) | 148-1, 151-1, 199-1 |
| ⊖ | (U+2296) | 149-1, 151-2 |
| ⊗ | (U+2297) | 64-1, 66-1, 69-1 |
| ⊘ | (U+2298) | 149-2, 152-1 |
| ⊙ | (U+2299) | 166-1 |
| ⊚ | (U+229A) | 87-3, 90-3, 93-1 |
| ⊛ | (U+229B) | 149-3, 151-3, 199-2 |
| ⊞ | (U+229E) | 80-2 |
| ⊠ | (U+22A0) | 19-3, 36-1 |
| ⊡ | (U+22A1) | 152-2 |
| ⊣ | (U+22A3) | 172-2 |
| ⊤ | (U+22A4) | 7-3, 200-1 |
| ⊢ | (U+22A6) | 200-2 |
| ⊶ | (U+22B0) | 90-4 |
| ⊲ | (U+22B2) | 148-2 |
| ⊳ | (U+22B3) | 148-3 |
| ⊷ | (U+22B6) | 21-3 |
| ⊸ | (U+22B7) | 110-3 |
| ⊹ | (U+22B8) | 93-2 |
| ⊹ | (U+22B9) | 134-1, 140-1 |
| ⋄ | (U+22C4) | 108-2 |
| ⋅ | (U+22C5) | 64-2, 66-2, 69-2 |
| ⋈ | (U+22C8) | 64-3, 66-3, 69-3 |
| ⋔ | (U+22D4) | 120-2 |
| ⋕ | (U+22D5) | 75-1 |
| ⋖ | (U+22D6) | 134-2, 140-2 |
| ⋘ | (U+22D8) | 148-4, 151-4 |
| ⋮ | (U+22EE) | 137-1, 140-3 |
| ⋯ | (U+22EF) | 138-1 |
| ⋲ | (U+22F2) | 78-1 |
| ⋳ | (U+22F3) | 78-2 |
| ⋵ | (U+22F5) | 103-3 |

| | | |
|---|---|---|
| ⋐ | (U+22F8) | 100-3 |
| ⋐ | (U+22F9) | 103-4 |
| ⋿ | (U+22FF) | 168-3 |
| ⌈ | (U+2308) | 25-1, 35-2 |
| ⌊ | (U+230A) | 25-2, 35-3 |
| ⌬ | (U+232C) | 117-2 |
| ⌼ | (U+233C) | 67-1 |
| ⌽ | (U+233D) | 78-3, 83-4 |
| ⌿ | (U+233F) | 107-3, 112-3 |
| ⍀ | (U+2340) | 108-3, 113-3 |
| ⍉ | (U+2349) | 78-4 |
| ⍋ | (U+234B) | 73-3 |
| ⍒ | (U+2352) | 73-4 |
| ⍗ | (U+2357) | 30-1 |
| ⍰ | (U+235B) | 18-1, 23-1, 33-1 |
| ⍠ | (U+2360) | 127-1 |
| ⍣ | (U+2363) | 122-1 |
| ⍳ | (U+2373) | 76-3 |
| ⍷ | (U+237C) | 28-1, 35-4 |
| ⎈ | (U+2388) | 80-3 |
| ⎊ | (U+238A) | 53-1 |
| ⎢ | (U+23A2) | 50-1, 53-2 |
| ⏀ | (U+23C0) | 51-1 |
| ⑳ | (U+2473) | 186-1, 233-1 |
| Ⓒ | (U+24B8) | 166-2 |
| Ⓘ | (U+24BE) | 166-3 |
| ▲ | (U+25B2) | 163-2 |
| ▼ | (U+25BC) | 163-3 |
| ● | (U+25CF) | 174-1 |
| ◝ | (U+25DD) | 25-3 |
| ◞ | (U+25DE) | 25-4 |
| ◶ | (U+25F6) | 26-1 |
| ◷ | (U+25F7) | 26-2 |
| ☘ | (U+2618) | 210-2, 210-3, 210-4, 211-2 |
| ☻ | (U+263B) | 177-1 |
| ⚁ | (U+2681) | 41-2 |
| ⚃ | (U+2683) | 41-3 |
| ⚅ | (U+2685) | 41-4 |
| ✔ | (U+2714) | 137-2 |
| ✧ | (U+2727) | 137-3 |
| ✪ | (U+272A) | 180-1, 220-1 |
| ❁ | (U+2741) | 173-1 |
| ❌ | (U+274C) | 134-3 |
| ❎ | (U+274E) | 134-4 |
| ❖ | (U+2756) | 186-2, 227-1 |
| ❗ | (U+2757) | 174-2 |
| ❶ | (U+2776) | 189-1, 220-2 |
| ❷ | (U+2777) | 189-2, 220-3 |
| ❸ | (U+2778) | 189-3, 220-4 |
| ❹ | (U+2779) | 190-1, 221-1 |
| ❺ | (U+277A) | 190-2, 221-2 |
| ❻ | (U+277B) | 180-2, 222-1 |
| ❼ | (U+277C) | 180-3, 236-2 |
| ❾ | (U+277E) | 190-3, 221-3 |
| ❿ | (U+277F) | 190-4, 221-4 |
| ⟒ | (U+27D2) | 51-2 |
| ⟚ | (U+27DA) | 82-1 |
| ⟰ | (U+27F0) | 169-1, 239-2 |
| ⟲ | (U+27F2) | 117-3 |
| ⟳ | (U+27F3) | 117-4 |
| ⤙ | (U+291A) | 120-3 |
| ⤺ | (U+293A) | 119-4 |
| ⦀ | (U+2980) | 53-3 |
| ⦉ | (U+2991) | 115-1 |
| ⦊ | (U+2992) | 115-2 |
| ⦙ | (U+2999) | 137-4 |
| ⦵ | (U+29B5) | 169-2, 238-3 |
| ⦼ | (U+29BD) | 79-4, 84-3 |
| ⧊ | (U+29CA) | 28-2 |
| ⧓ | (U+29D3) | 169-3, 238-4 |
| ⧗ | (U+29D7) | 169-4, 239-3 |
| ⧾ | (U+29DE) | 115-3 |
| ⨂ | (U+2A02) | 18-2, 23-2, 33-2 |
| ⨃ | (U+2A03) | 42-1, 44-2 |
| ⨊ | (U+2A0A) | 75-2, 82-2 |
| ⨢ | (U+2A22) | 17-3, 22-3, 32-3 |
| ⨣ | (U+2A23) | 76-4 |
| ⨧ | (U+2A27) | 21-4 |
| ⨬ | (U+2A2C) | 18-3, 23-3, 33-3 |
| ⨯ | (U+2A2F) | 19-4, 24-4, 34-4 |
| ⨰ | (U+2A30) | 17-4, 22-4, 32-4 |
| ⨳ | (U+2A33) | 127-2 |
| ⨷ | (U+2A37) | 89-3, 93-3, 199-3 |
| ⨸ | (U+2A38) | 18-4, 23-4, 33-4, 199-4 |

| | | |
|---|---|---|
| ⩒ | (U+2A52) | 100-4 |
| ⩓ | (U+2A53) | 104-4 |
| ⩔ | (U+2A54) | 28-3 |
| ⩕ | (U+2A55) | 200-3 |
| ⩖ | (U+2A56) | 200-4 |
| ⩞ | (U+2A5E) | 75-3, 82-3 |
| ⩟ | (U+2A5F) | 29-4 |
| ⩩ | (U+2A69) | 51-3, 54-1 |
| ⩸ | (U+2A78) | 110-4 |
| ⪦ | (U+2AA6) | 27-1 |
| ⪧ | (U+2AA7) | 27-2 |
| ⪭ | (U+2ABD) | 27-3 |
| ⪮ | (U+2ABE) | 28-4 |
| ⫍ | (U+2ACD) | 127-3 |
| ⫎ | (U+2ACE) | 127-4 |
| ⫏ | (U+2ACF) | 108-4, 115-4 |
| ⫑ | (U+2AD1) | 27-4 |
| ⫗ | (U+2AD7) | 26-3 |
| ⫘ | (U+2AD8) | 26-4 |
| ⫩ | (U+2AE9) | 93-4 |
| ⫮ | (U+2AEE) | 50-2, 53-4 |
| ⫵ | (U+2AF5) | 89-4 |
| ⫷ | (U+2AF7) | 113-4, 122-2 |
| ⫽ | (U+2AFD) | 60-3 |
| ⬄ | (U+2B04) | 152-3 |
| ⸙ | (U+2E19) | 211-3 |
| 支 | (U+2F40) | 211-4 |
| ㉈ | (U+3248) | 193-1, 235-1 |
| ㉉ | (U+3249) | 194-1, 235-2 |
| ㉇ | (U+3267) | 193-2, 230-1 |
| ㈠ | (U+3280) | 181-1, 222-2 |
| ㈡ | (U+3281) | 181-2, 222-3 |
| ㈢ | (U+3282) | 181-3, 223-1 |
| ㈣ | (U+3283) | 181-4, 223-2 |
| ㈤ | (U+3284) | 182-1, 223-3 |
| ㈥ | (U+3285) | 182-2, 223-4 |
| ㈦ | (U+3286) | 182-3, 224-1 |
| ㈧ | (U+3287) | 182-4, 224-2 |
| ㈨ | (U+3288) | 183-1, 224-3 |
| ㈩ | (U+3289) | 183-2, 224-4 |
| ㈪ | (U+328A) | 183-3, 225-1 |
| ㈬ | (U+328C) | 194-2, 228-1 |
| ㈭ | (U+328D) | 194-3, 228-2 |
| ㈮ | (U+328E) | 194-4, 228-3 |
| ㈯ | (U+328F) | 195-1, 228-4 |
| ㈰ | (U+3290) | 195-2, 229-1 |
| ㉿ | (U+3297) | 180-4, 222-4 |
| ㊣ | (U+329F) | 196-1, 230-2 |
| ㊠ | (U+32A0) | 196-2, 230-3 |
| ㊲ | (U+32A2) | 196-3, 230-4 |
| ㊱ | (U+32A9) | 195-3, 229-2 |
| ㊷ | (U+32AA) | 195-4, 229-3 |
| ㊸ | (U+32AB) | 196-4, 229-4 |
| 丽 | (U+4E3D) | 187-2, 236-3, 239-4 |
| 侠 | (U+4FA0) | 7-1 |
| ꝋ | (U+A74B) | 187-3, 236-4 |
| ꭠ | (U+AB44) | 61-1 |
| 𝐂 | (U+1D21D) | 155-2, 158-1 |
| 𝒜 | (U+1D49C) | 75-4, 82-4 |
| 𝒞 | (U+1D49E) | 130-2 |
| 𝒫 | (U+1D4AB) | 50-3 |
| 𝒱 | (U+1D4E5) | 155-3 |
| 𝔏 | (U+1D50F) | 58-3 |
| 𝔓 | (U+1D513) | 50-4 |
| 𝔸 | (U+1D538) | 172-3 |
| 𝔽 | (U+1D53D) | 173-2 |
| 𝕀 | (U+1D540) | 166-4 |
| 𝕄 | (U+1D544) | 177-2 |
| 𝕆 | (U+1D546) | 168-4 |
| 𝕊 | (U+1D54A) | 173-3 |
| 𝕋 | (U+1D54B) | 173-4 |
| 𝕎 | (U+1D54E) | 172-4 |
| 𝕏 | (U+1D54F) | 72-3 |
| 𝕐 | (U+1D550) | 72-4 |
| 𝔅 | (U+1D56D) | 39-3, 44-3, 47-1 |
| 𝔇 | (U+1D56F) | 40-4, 44-4, 47-2 |
| 𝔍 | (U+1D574) | 107-4 |
| 𝔏 | (U+1D577) | 66-4 |
| 𝔑 | (U+1D57D) | 39-4 |
| 𝔲 | (U+1D580) | 67-2 |
| 𝔷 | (U+1D585) | 112-4 |
| 𝖋 | (U+1D58B) | 60-4 |
| 𝟘 | (U+1D7D8) | 155-4 |
| 𝟙 | (U+1D7D9) | 156-1 |

| | | |
|---|---|---|
| 𝟚 | (U+1D7DA) | 156-2 |
| 𝟛 | (U+1D7DB) | 156-3 |
|  | (U+1F000) | 188-1, 234-1 |
|  | (U+1F001) | 188-2, 234-2 |
|  | (U+1F002) | 188-3, 234-3 |
|  | (U+1F003) | 188-4, 235-3 |
|  | (U+1F029) | 189-4, 235-4 |
|  | (U+1F02A) | 187-4, 234-4 |
|  | (U+1F030) | 186-3, 231-1 |
|  | (U+1F032) | 191-1, 231-2 |
|  | (U+1F033) | 191-2, 231-3 |
|  | (U+1F034) | 191-3, 231-4 |
|  | (U+1F035) | 191-4, 232-1 |
|  | (U+1F036) | 192-1, 232-2 |
|  | (U+1F037) | 192-2, 232-3 |
|  | (U+1F038) | 192-3, 232-4 |
|  | (U+1F039) | 192-4, 233-2 |
|  | (U+1F03A) | 193-3, 233-3 |
|  | (U+1F03B) | 193-4, 233-4 |
| 手 | (U+1F210) | 183-4, 225-2 |
| 字 | (U+1F211) | 184-1, 225-3 |
| 双 | (U+1F212) | 184-2, 225-4 |
| デ | (U+1F213) | 184-3, 226-1 |
| 二 | (U+1F214) | 184-4, 226-2 |
| 多 | (U+1F215) | 185-1, 226-3 |
| 解 | (U+1F216) | 185-2, 226-4 |
| 天 | (U+1F217) | 185-3, 227-2 |
| 交 | (U+1F218) | 185-4, 227-3 |
| 映 | (U+1F219) | 186-4, 227-4 |